

1. Rice

Rice (*Oryza sativa*) belongs to the Poaceae (grass) family and is one of the most important cereal food crops in the world. It feeds more than half of the global population and forms the primary staple diet in countries such as India, China, Bangladesh, Indonesia, and Vietnam. In India, rice is mainly cultivated during the Kharif season (June–October) under monsoon rainfall, and in irrigated regions it is also grown during Rabi and summer seasons. Rice cultivation supports millions of farmers and agricultural laborers, playing a crucial role in food security, rural employment, and the national economy. Beyond direct consumption, rice is used in processed foods, rice bran oil production, brewing industries, and as animal feed.

Rice requires warm and humid climatic conditions. For proper germination, temperatures between 20–25°C are ideal, while active vegetative growth occurs best at 25–35°C. The flowering stage is particularly sensitive to temperature stress; extreme heat or cold can reduce grain filling and yield. Rice generally needs 1000–2000 mm of annual rainfall and can be cultivated under both irrigated and rainfed systems. Soils with good water retention capacity, such as clay loam to loamy soils, are most suitable. The ideal soil pH ranges from 5.5 to 7.5. In India, major rice-growing states include West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Telangana, Tamil Nadu, Odisha, and Chhattisgarh.

Rice production is significantly affected by diseases and pests, which can cause severe yield losses if not managed properly. One of the most destructive diseases is Rice Blast, caused by the fungus *Magnaporthe oryzae*. It appears as diamond-shaped lesions with gray centers and brown margins on leaves. In severe cases, leaves dry completely, and during the reproductive stage, neck rot occurs, leading to poor grain filling. This disease thrives under high humidity, low night temperatures, and excessive nitrogen fertilizer application. If unmanaged, blast can cause yield losses of 50–70%. Prevention includes using resistant varieties, maintaining proper plant spacing for air circulation, avoiding excess nitrogen, and keeping the field clean. Seed treatment with bioagents such as *Pseudomonas fluorescens* helps reduce early infection. Chemical control through spraying Tricyclazole or Isoprothiolane is recommended at the early tillering stage and repeated after 10–15 days if symptoms persist. Continuous monitoring for at least 30–40 days after transplanting is critical, as early-stage infection spreads rapidly under favorable conditions.

Bacterial Leaf Blight (BLB), caused by *Xanthomonas oryzae*, is another serious disease. It begins with yellowing from the leaf tip downward, followed by water-soaked streaks that later dry and wilt. In severe infections, a condition called “Kresiek” leads to complete wilting of young plants. This disease spreads rapidly during heavy rainfall and strong winds, especially when excessive nitrogen is applied. Yield losses may range from 20–60%. Preventive measures include using certified disease-resistant seeds, balanced fertilizer application, proper drainage to avoid water stagnation, and field sanitation. Spraying Streptocycline (0.01%) combined with Copper Oxychloride controls bacterial spread. Organic options such as neem extract sprays can also reduce infection. Once symptoms appear, spraying should be done immediately and

repeated after 7–10 days. Monitoring is especially important within the first 40–50 days after transplanting, as early infection is more destructive.

Sheath Blight, caused by *Rhizoctonia solani*, is a fungal disease common in densely planted fields with high humidity and excess nitrogen. It produces oval greenish-gray lesions on the leaf sheath that spread upward, sometimes causing lodging. Yield loss can range from 10–40%. Prevention focuses on proper plant spacing, moderate nitrogen use, and removal of crop residues after harvest. Soil application of *Trichoderma* during land preparation reduces fungal inoculum. If disease symptoms are visible, spraying Validamycin or Hexaconazole is effective, repeated after 10–12 days if needed. Vigilance is required from 45–70 days after planting, as the disease intensifies during the vegetative to reproductive stages.

Brown Spot, caused by *Bipolaris oryzae*, produces circular brown lesions on leaves and can lead to seed discoloration and poor grain quality. It is more common in nutrient-deficient soils, particularly where potassium is lacking. Seed treatment with fungicides or bioagents helps prevent early infection. Balanced fertilization, especially adequate potassium, strengthens plant resistance. Spraying Mancozeb at symptom appearance, followed by another spray after 10–14 days, effectively controls spread. Compost tea sprays can be used as an organic alternative. Fields should be carefully observed from 30 days onward to prevent spread during humid conditions.

Tungro Disease is a viral disease transmitted by the green leafhopper. Infected plants show yellow-orange discoloration, stunted growth, and reduced tillering. In severe outbreaks, yield losses may reach up to 80%. There is no direct cure once plants are infected, so prevention is crucial. Managing the leafhopper vector through timely spraying of Imidacloprid, installing yellow sticky traps, and using neem oil sprays can significantly reduce transmission. Early planting and resistant varieties further minimize risk. Farmers must monitor fields regularly, particularly within 20–45 days after transplanting, as early infection leads to maximum damage.

In addition to diseases, pests like Stem Borer and Brown Planthopper (BPH) cause major losses. Stem borer larvae cause “dead heart” during the vegetative stage and “white ear heads” during the reproductive stage. Pheromone traps help in early detection. Chlorantraniliprole spraying is effective when infestation is noticed, usually 25–45 days after planting. Neem cake application in soil offers organic protection. Brown Planthopper causes hopper burn, where plants turn brown and dry. Avoiding excess nitrogen and ensuring good drainage helps prevent outbreaks. Spraying Imidacloprid or Buprofezin at early detection and repeating after 10 days if necessary is recommended.

Nutrient deficiencies also reduce crop productivity. Nitrogen deficiency causes yellowing of older leaves, phosphorus deficiency leads to stunted growth with dark green coloration, potassium deficiency causes brown leaf margins, and zinc deficiency results in white stripes known as Khaira disease. Applying recommended fertilizers based on soil testing is essential. Zinc sulfate application corrects zinc deficiency within 7–14 days of treatment.

General preventive farming practices include using certified disease-free seeds, crop rotation to break disease cycles, maintaining recommended plant spacing, balanced fertilizer use, proper irrigation management, avoiding continuous water stagnation, and regular field monitoring. Organic management practices such as seed treatment with *Trichoderma*, neem oil sprays (3%), Panchagavya application, Jeevamrutham soil treatment, and biofertilizers strengthen plant immunity and reduce dependency on chemicals.

Different regions in India adopt specialized practices. Punjab and Haryana use laser land leveling and controlled irrigation to conserve water. Andhra Pradesh and Telangana promote the System of Rice Intensification (SRI) to improve yield with less water. Tamil Nadu follows Alternate Wetting and Drying (AWD) irrigation methods. Eastern India increasingly adopts Direct Seeded Rice (DSR) to reduce labor and water usage.

Effective disease prevention depends on early detection, timely spraying, balanced nutrition, and integrated pest management. Most fungal diseases require monitoring from 20 to 70 days after planting, while bacterial and viral diseases need early-stage prevention within the first 40 days. Timely action within 7–15 days of symptom appearance significantly reduces spread. With proper preventive care and regular monitoring, farmers can minimize yield losses and maintain healthy rice production.

2. Gram

Gram, commonly known as chickpea or Bengal gram, scientifically named *Cicer arietinum*, belongs to the Fabaceae (legume) family. It is one of the most important pulse crops grown in India and other semi-arid regions of the world. India is the largest producer and consumer of chickpea, and the crop is mainly cultivated during the Rabi season (October–March), mostly under rainfed conditions. Chickpea is highly valued for its nutritional importance and soil-enriching properties. The seeds contain about 18–22% protein, along with carbohydrates, dietary fiber, calcium, iron, and essential vitamins, making it a major protein source in vegetarian diets. It is used in the preparation of dal (split gram), besan (gram flour), snacks, processed foods, and also as animal feed. Being a leguminous crop, it fixes atmospheric nitrogen in the soil through symbiotic Rhizobium bacteria, thereby improving soil fertility and reducing fertilizer requirements for subsequent crops. It requires comparatively less water than cereals and is highly suitable for crop rotation systems, providing stable income for small and marginal farmers.

Chickpea grows best under cool and dry climatic conditions. Germination occurs optimally at temperatures between 20–25°C, while vegetative growth performs well between 18–30°C. The crop is sensitive to frost, excessive humidity, and continuous rainfall, especially during flowering and pod formation stages, which can severely reduce yield. It requires an annual rainfall of 400–700 mm and performs well under residual soil moisture in rainfed areas. Bright sunshine during pod development improves grain filling and final yield. Chickpea grows well in well-drained sandy loam to clay loam soils with a pH range of 6.0–8.0. Deep soils are preferred to allow proper root penetration. Waterlogging and saline soils must be strictly avoided, as excess moisture predisposes the crop to root diseases.

Major gram-producing states in India include Madhya Pradesh (the largest producer), Maharashtra, Rajasthan, Uttar Pradesh, Andhra Pradesh, Telangana, Karnataka, Gujarat, and Chhattisgarh. Chickpea varieties are broadly classified into Desi and Kabuli types. Desi types have small, dark-colored seeds and are more drought tolerant, while Kabuli types have large, cream-colored seeds with higher market value. Popular varieties include JG 11, JG 14, Pusa 372, Pusa 256, KAK 2, and ICCV series.

Chickpea productivity is significantly affected by diseases, particularly fungal and viral infections. Fusarium wilt, caused by *Fusarium oxysporum* f.sp. *ciceri*, is one of the most destructive diseases, capable of causing yield losses up to 60%. It can occur at any growth stage. Symptoms include yellowing of leaves, drooping, sudden wilting, and eventual plant death. When the stem is cut open, brown discoloration of vascular tissues is visible. The disease is favored by warm soil temperatures, continuous chickpea cultivation, and poor drainage. Prevention begins before sowing by adopting crop rotation for at least 3–4 years with non-host crops such as cereals. Deep summer ploughing exposes fungal spores to sunlight and reduces inoculum load. Use of resistant varieties and certified disease-free seeds is essential. Seed treatment with Carbendazim or Thiram should be done 1–2 days before sowing. Organic

farmers can treat seeds with *Trichoderma viride*, which protects roots for approximately 30–40 days after germination. Fields should be closely monitored from 20 days after sowing up to 60 days, as early-stage infection spreads quickly. Once symptoms appear, affected plants should be removed immediately to prevent spread, although complete cure is not possible at advanced stages.

Ascochyta blight, caused by *Ascochyta rabiei*, can cause 50–70% yield loss in cool and humid conditions. It produces brown lesions with dark margins on leaves, stems, and pods. Severe infection leads to drying of plants. The disease spreads rapidly during continuous rainfall and cloudy weather. Preventive strategies include using certified seeds, avoiding dense sowing to improve air circulation, and maintaining strict field sanitation. Fungicidal sprays of Mancozeb or Chlorothalonil should be applied at the first appearance of spots and repeated after 10–12 days if humid conditions persist. Protection should be maintained especially between 30–70 days after sowing, which is the most vulnerable period. Organic control through neem-based fungicides provides partial suppression and should be repeated at 10-day intervals during high-risk weather.

Botrytis Gray Mold appears as gray fungal growth on leaves and flowers, causing flower drop and poor pod formation under high humidity and cloudy conditions. Preventive measures include wider spacing, timely sowing to avoid peak humidity periods, and ensuring good air movement within the crop canopy. Carbendazim spray at early symptom detection followed by another spray after 10 days provides control. Monitoring is critical during flowering (45–75 days after sowing), as prolonged moisture for more than 3–4 consecutive days increases risk.

Root rot causes sudden drying and decay of roots, mainly in poorly drained soils. Prevention focuses on well-drained land, crop rotation, and seed treatment with fungicides or *Trichoderma*. Fields should be monitored during the vegetative stage (20–50 days). Avoid irrigation stagnation for more than 24–48 hours to reduce infection risk.

Chickpea Mosaic Virus leads to yellow mosaic patterns, stunted growth, and reduced yield (20–50% loss). It is transmitted by aphids. There is no direct cure once infected; therefore, vector control is essential. Spray Imidacloprid at early aphid detection and repeat after 10–15 days if required. Neem oil sprays at 7–10 day intervals help reduce aphid populations organically. The most critical prevention window is the first 40 days after sowing, as early infection severely reduces yield.

Among pests, Pod Borer (*Helicoverpa armigera*) is the most damaging, causing 30–80% yield loss. Larvae bore into pods and feed on developing seeds. Installation of pheromone traps should begin 25–30 days after sowing for early detection. Spraying Spinosad or Emamectin should be done at flowering and repeated after 10–12 days if infestation continues. Organic control includes neem oil sprays every 7–10 days and releasing *Trichogramma* parasitoids. Aphids suck plant sap and cause leaf curling; early spraying of Imidacloprid or neem oil at 10-day intervals prevents heavy infestation. Cutworms cut young seedlings at ground level; prevention includes proper field sanitation and soil treatment before sowing.

Nutrient deficiencies also affect plant growth. Nitrogen deficiency causes pale green leaves, phosphorus deficiency leads to poor root growth, potassium deficiency results in weak plants, and zinc deficiency causes stunted growth. Though chickpea fixes nitrogen, seed inoculation with *Rhizobium* culture ensures effective nodulation within 20–25 days after sowing. Balanced fertilizer application based on soil testing corrects deficiencies.

Proper seed treatment is essential for early disease prevention. Seeds should first be treated with fungicides, then inoculated with *Rhizobium* culture, and finally coated with *Trichoderma* for biological protection. This combined protection remains effective for about 30–45 days during early crop establishment.

General preventive farming practices include timely sowing during October–November, maintaining recommended spacing, effective weed management within the first 30 days, avoiding waterlogging, practicing crop rotation, and regular field monitoring every 7–10 days. Chickpea is mostly rainfed, but irrigation at critical stages such as flowering and pod formation improves yield. Excess irrigation must be avoided, especially during vegetative growth.

Organic farming practices involve seed treatment with *Trichoderma*, application of neem cake in soil, use of Jeevamrutham and Panchagavya for enhancing soil microbial activity, and biological pest control methods. Compost and farmyard manure improve soil structure and disease resistance.

Harvesting should be done when plants turn yellow and dry completely. Delayed harvesting may lead to pod shattering. After harvest, seeds must be properly dried to reduce moisture below safe storage levels and stored in dry, pest-free conditions.

Region-specific practices in India include rainfed improved varieties in Madhya Pradesh, drought-tolerant varieties in Rajasthan, residual moisture cultivation after rice in Telangana and Andhra Pradesh, and integrated pest management (IPM) in Maharashtra.

In summary, most chickpea diseases require preventive management before sowing and vigilant monitoring from 20 to 70 days after sowing. Fungal diseases require protection sprays at 10–12 day intervals during favorable weather. Viral diseases require strict vector control within the first 40 days. Early detection and timely intervention can reduce yield losses significantly and ensure healthy crop production.

3. Moong

Moong, commonly known as Green Gram, scientifically called *Vigna radiata*, belongs to the Fabaceae family. It is a short-duration pulse crop with a crop cycle of about 60–75 days, making it highly suitable for multiple cropping systems. It is widely cultivated in tropical and subtropical regions and is one of the major pulse crops in India. Moong can be grown during Kharif (June–July), Rabi in South India (October–November), and Summer season (March–April). The crop has a deep root system with nitrogen-fixing nodules that enrich soil fertility naturally by fixing about 25–30 kg nitrogen per hectare. Due to its short duration and soil-improving nature, it is ideal for crop rotation, intercropping, and rice-fallow systems. Economically, it provides quick returns with low input cost and strong market demand, benefiting small and marginal farmers.

Nutritionally, moong contains about 22–24% protein and is rich in carbohydrates, fiber, iron, calcium, and potassium. It is easily digestible compared to other pulses and is used for dal preparation, sprouts, baby food, flour, and snack products. Agriculturally, it improves soil structure, reduces fertilizer needs for subsequent crops, and helps break pest and disease cycles.

Moong requires a warm climate with an optimum temperature range of 25–35°C for growth. Germination can occur at a minimum temperature of 15°C. The crop is sensitive to frost and highly sensitive to waterlogging. Bright sunshine during flowering and pod development improves yield, while cloudy and humid weather increases disease incidence. It requires about 500–700 mm rainfall annually. Excess rainfall during flowering or pod formation reduces yield significantly.

The crop grows best in sandy loam to loamy soils with good drainage and aeration. Ideal soil pH ranges from 6.0 to 7.5. Heavy clay, saline, alkaline soils, and waterlogged conditions must be avoided, as root respiration is severely affected. Major moong-growing states in India include Rajasthan, Maharashtra, Madhya Pradesh, Uttar Pradesh, Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Odisha, and Punjab and Haryana (mainly as a summer crop).

Land preparation involves one deep ploughing followed by one or two harrowings to create a fine seedbed. Removal of crop residues and proper leveling ensures uniform irrigation. Summer ploughing helps reduce pest and disease inoculum by exposing soil-borne pathogens to sunlight.

Sowing should be done during June–July for Kharif, March–April for Summer, and October–November in South India for Rabi. A seed rate of 12–15 kg per hectare is recommended, with spacing of 30 cm between rows and 10 cm between plants. Seeds should be placed 3–5 cm deep. Line sowing is preferred over broadcasting to ensure proper spacing and aeration. Improved varieties such as Pusa Vishal, Pusa Ratna, SML 668, IPM 02-3, LGG 460, WGG series, Pant Moong varieties, and Yellow Mosaic Virus (YMV) resistant varieties are recommended for higher yield and disease tolerance.

Seed treatment is a critical preventive step. Seeds should be treated with Carbendazim or Thiram to protect against seed-borne fungal infections. After fungicide treatment, inoculation with *Rhizobium* culture enhances nitrogen fixation, and *Trichoderma* treatment protects against soil-borne pathogens. This protection generally remains effective for 30–40 days after germination.

Nutrient management includes application of 15–20 kg nitrogen and 40–50 kg phosphorus per hectare as basal dose. Potassium at 20 kg/ha should be applied in deficient soils. Micronutrient foliar sprays can be given if deficiency symptoms appear. Biofertilizers such as *Rhizobium* and Phosphate Solubilizing Bacteria (PSB) improve nutrient availability.

Moong is mostly grown under rainfed conditions. However, irrigation at critical stages such as flowering (30–35 days after sowing) and pod formation (45–50 days) significantly increases yield. Excess irrigation and water stagnation for more than 24 hours must be avoided to prevent root diseases.

Weed management is essential during the first 20–30 days, which is the critical competition period. Hand weeding at 20 days and again at 35 days after sowing is effective. Pre-emergence application of Pendimethalin controls early weeds. Organic farmers can use mulching.

Among diseases, Yellow Mosaic Virus (YMV) is the most destructive, causing up to 80% yield loss. It is transmitted by whitefly. Symptoms include yellow mosaic patches, leaf distortion, stunted growth, and poor pod formation. Warm and humid conditions favor whitefly multiplication. Prevention should start immediately after germination and continue up to 40 days after sowing, as early infection causes maximum loss. Resistant varieties should be used. Infected plants must be removed immediately to prevent spread. Imidacloprid or Thiamethoxam spraying at early whitefly detection should be done and repeated after 10–12 days if needed. Organic control includes neem oil (3%) spray at 7–10 day intervals and installation of yellow sticky traps throughout the first 45 days.

Cercospora Leaf Spot causes brown circular lesions that enlarge and merge, leading to leaf drying and drop. It is common under high humidity and rainy conditions. Protection should begin when the crop is 25–30 days old and continue during humid weather. Spray Mancozeb at first symptom appearance and repeat after 10–14 days. Organic neem extract sprays at similar intervals help reduce infection. Monitoring should be done from 30–60 days after sowing.

Powdery Mildew appears as white powdery growth on leaves during dry weather with cool nights. Leaves turn yellow and dry. Spray Wettable Sulfur at early symptom detection and repeat after 10 days if needed. Organic control can be done with baking soda mixed with neem oil, applied every 7–10 days during favorable conditions.

Anthrachnose causes dark lesions on stems and pods, leading to pod shriveling. Seed treatment prevents early infection. Spray Carbendazim at symptom appearance and repeat after 10–12 days. Monitoring is important during flowering and pod development stages (35–60 days).

Root Rot causes yellowing, wilting, root decay, and plant death, particularly in poorly drained soils. Prevention includes proper drainage and seed treatment with *Trichoderma*. Avoid water stagnation beyond 24 hours. The most critical period is 20–50 days after sowing.

Among insect pests, Pod Borer is highly damaging, causing 30–60% yield loss. It creates holes in pods and feeds on seeds. Install pheromone traps from 25 days after sowing for early monitoring. Spray Spinosad at flowering and repeat after 10–12 days if infestation persists. Neem oil sprays at 7-day intervals and biological control agents can be used in organic systems.

Whiteflies not only suck sap but also transmit YMV. Aphids cause leaf curling and sap loss. Thrips damage flowers, reducing pod formation. Imidacloprid or neem oil sprays at early detection and repeated every 10 days during heavy infestation provide control.

Integrated Pest Management (IPM) includes using resistant varieties, crop rotation, field sanitation, biological control agents, pheromone traps, and need-based pesticide application. Regular field monitoring every 5–7 days is essential, especially between 20–50 days after sowing.

Organic farming practices involve seed treatment with *Trichoderma*, neem cake application, Panchagavya spray, Jeevamrutham soil application, vermicompost use, and biological pest control methods.

Nutrient deficiencies manifest as pale green leaves (nitrogen), poor root growth (phosphorus), weak plants (potassium), and stunted growth (zinc). Balanced fertilization and micronutrient sprays correct these within 7–14 days after application.

Harvesting should be done when about 80% of pods turn dark and mature. Delayed harvesting increases pod shattering losses. After harvest, seeds must be dried to 10–12% moisture and stored in clean, dry containers. Neem leaves can be mixed with stored grains for natural pest protection.

Average yield ranges from 8–12 quintals per hectare under normal practices and can reach 15–18 quintals per hectare with improved management. Major yield losses are caused by YMV (up to 80%), leaf spot (30–40%), pod borer (30–60%), and root rot (20–30%). Most diseases require preventive monitoring from 20–60 days after sowing, with spray intervals of 7–14 days during favorable conditions.

Common cropping systems include Rice–Moong, Wheat–Summer Moong, and intercropping with cotton, maize, and sugarcane. Regional practices vary: Punjab and Haryana grow summer moong after wheat; Andhra Pradesh and Telangana cultivate rice-fallow moong; Rajasthan practices rainfed Kharif cultivation; Maharashtra adopts intercropping systems.

Effective prevention relies on early sowing, resistant varieties, seed treatment, proper drainage, vector control within the first 40 days, and regular monitoring. Timely action within 7–10 days of symptom appearance significantly reduces disease spread and protects yield.

4. Urad

Urad, commonly known as Black gram, Urad dal, or Mash, scientifically named *Vigna mungo*, belongs to the Fabaceae family. It is one of the most important pulse crops grown in India and plays a major role in food security and soil fertility improvement. Urad is a short-duration legume crop with a life cycle of about 70–90 days. It is cultivated during Kharif (June–July), Rabi in South India (October–November), and Summer season (February–March). The plant is bushy and erect or semi-spreading in nature, with a well-developed root system containing nitrogen-fixing nodules that enrich the soil by fixing around 20–25 kg nitrogen per hectare. It can be grown under both rainfed and irrigated conditions and is widely used in Indian cuisine for dal, idli and dosa batter, papad, snacks, and sweets.

Nutritionally, urad contains 24–26% protein and is rich in carbohydrates, iron, calcium, magnesium, potassium, and dietary fiber. It is easily digestible when properly cooked and is an essential component of vegetarian diets. Agriculturally, urad improves soil structure, reduces fertilizer requirements for subsequent crops, and fits well into crop rotation and intercropping systems, making it important for sustainable farming. Economically, it offers high market demand, low cultivation cost, and quick returns due to its short duration, benefiting small and marginal farmers.

Urad requires a warm and humid climate with an optimal temperature range of 25–35°C. It is sensitive to frost and highly sensitive to waterlogging. The crop requires about 600–800 mm rainfall annually. Excess rainfall during flowering can cause flower drop and reduced yield, while dry weather during maturity is essential for proper pod drying. Bright sunlight during pod formation ensures better grain filling.

The crop performs best in well-drained loamy or sandy loam soils with good aeration. The ideal soil pH is between 6.0 and 7.5. Saline, alkaline, heavy clay soils and poorly drained fields should be avoided to prevent root diseases. Major urad-growing states in India include Uttar Pradesh, Madhya Pradesh, Maharashtra, Andhra Pradesh, Telangana, Tamil Nadu, Rajasthan, Bihar, Karnataka, and West Bengal.

Land preparation involves one deep ploughing followed by two harrowings to create a fine tilth. Crop residues and weeds should be removed. Proper leveling ensures uniform irrigation. Summer ploughing reduces pest and disease inoculum by exposing soil-borne pathogens to sunlight.

Sowing is done during June–July for Kharif, October–November for Rabi in South India, and February–March for Summer. A seed rate of 15–20 kg per hectare is recommended, with 30 cm row spacing and 10 cm plant spacing. Seeds should be sown at a depth of 3–5 cm using line sowing to maintain uniform plant population. Improved varieties such as PU-31, LBG-752, T-9, Pant Urd, and IPU series are recommended, especially Yellow Mosaic Virus (YMV) resistant varieties.

Seed treatment is essential to prevent early-stage diseases. Seeds should first be treated with Carbendazim or Thiram to control seed-borne fungi. After fungicide treatment, seeds should be inoculated with *Rhizobium* culture to enhance nitrogen fixation and treated with *Trichoderma* for protection against soil-borne pathogens. This protective effect generally lasts for about 30–40 days after sowing.

Nutrient management includes applying 15–20 kg nitrogen and 40–50 kg phosphorus per hectare as basal dose. Potassium at 20 kg/ha may be applied in deficient soils. Farmyard manure or compost should be incorporated before sowing. Biofertilizers such as *Rhizobium* and Phosphate Solubilizing Bacteria improve nutrient availability.

Urad is mostly rainfed, but irrigation at critical stages—flowering (35–40 days after sowing) and pod formation (45–55 days)—increases yield. Excess irrigation must be avoided. Water stagnation beyond 24 hours increases root disease risk.

Weed control is critical during the first 30 days after sowing. Hand weeding at 20–25 days and again at 35 days ensures good crop growth. Pre-emergence application of Pendimethalin controls early weeds.

Among diseases, Yellow Mosaic Virus (YMV) is the most destructive, causing yield losses up to 85%. It is transmitted by whitefly. Symptoms include yellow mosaic patches on leaves, stunted growth, and poor pod formation. Warm and humid weather increases whitefly multiplication. Prevention must focus on the first 40 days after sowing, as early infection causes maximum damage. Resistant varieties should be selected. Infected plants should be removed immediately to prevent spread. Spray Imidacloprid or Thiamethoxam at early whitefly detection and repeat after 10–12 days if necessary. Organic control includes neem oil sprays at 7–10 day intervals and installation of yellow sticky traps throughout the initial 45 days.

Cercospora Leaf Spot appears as brown circular spots that enlarge and merge, causing premature leaf fall. It thrives in humid and rainy conditions. Monitoring should begin around 25–30 days after sowing. Spray Mancozeb or Carbendazim at the first appearance of spots and repeat after 10–14 days if humidity persists. Field sanitation and removal of infected debris reduce disease spread.

Powdery Mildew develops as white powdery growth on leaves during dry weather with cool nights. It causes leaf drying and reduced photosynthesis. Spray Wettable Sulfur at early symptom appearance and repeat after 10 days if required. Neem oil can be used as an organic alternative.

Anthrachnose causes dark lesions on stems and pods, leading to pod shriveling and reduced grain quality. Proper seed treatment prevents early infection. Spray Carbendazim at symptom detection and repeat after 10–12 days if necessary. Monitoring is essential between 35–60 days after sowing.

Root Rot results in yellowing, wilting, root decay, and plant death, particularly in poorly drained soils. Prevention includes proper drainage and seed treatment with *Trichoderma*. Avoid water

stagnation for more than 24 hours. The most vulnerable period is between 20–50 days after sowing.

Major insect pests include Pod Borer, Whitefly, Aphids, and Thrips. Pod borer damages pods and seeds, causing 30–60% yield loss. Install pheromone traps from 25 days after sowing for early detection. Spray Spinosad at flowering and repeat after 10–12 days if infestation continues. Neem oil sprays at 7-day intervals are effective in organic systems. Whiteflies suck sap and transmit YMV, requiring early insecticide or neem spray intervention. Aphids cause leaf curling and reduced vigor; systemic insecticides or neem extract control them. Thrips damage flowers, reducing pod formation.

Integrated Pest Management (IPM) combines resistant varieties, crop rotation, field sanitation, biological control agents, pheromone traps, and need-based pesticide application. Regular field monitoring every 5–7 days, especially between 20–60 days after sowing, is essential.

Organic farming practices include seed treatment with *Trichoderma*, neem cake application, vermicompost use, Panchagavya sprays, Jeevamrutham soil treatment, and neem oil for pest control.

Nutrient deficiencies show as pale leaves (nitrogen), poor root growth (phosphorus), weak plants (potassium), and interveinal yellowing (iron). Balanced fertilization and foliar micronutrient sprays correct deficiencies within 7–14 days.

Harvesting should be done when 80–85% of pods mature and turn black. Delayed harvesting leads to pod shattering losses. Plants should be cut and dried before threshing. After harvest, seeds must be dried to 10–12% moisture and stored in airtight containers. Neem leaves can be used for storage pest protection.

Average yield ranges from 8–10 quintals per hectare under normal conditions and 12–15 quintals with improved practices. Under ideal conditions with high-yielding varieties and proper irrigation, yields can reach up to 18 quintals per hectare. Major yield losses occur due to YMV (up to 85%), leaf spot (30–40%), pod borer (30–60%), and root rot (20–30%).

Urad fits well in cropping systems such as Rice–Urad, Wheat–Summer Urad, and intercropping with maize, cotton, and sugarcane. Regionally, Punjab and Haryana cultivate summer urad after wheat, Andhra Pradesh and Telangana practice rice-fallow cultivation, Rajasthan and Madhya Pradesh grow rainfed Kharif crops, and Tamil Nadu and Karnataka cultivate Rabi crops.

Urad is relatively drought tolerant and short in duration, making it suitable for climate-resilient and climate-smart agriculture. Most disease prevention measures should focus on the first 40–60 days after sowing, with spray intervals of 7–14 days during favorable conditions. Early detection and timely control significantly reduce yield loss and ensure healthy crop production.

5. Arhar

Arhar, also known as Tur or Pigeon Pea, scientifically named *Cajanus cajan*, belongs to the Fabaceae (legume) family and is one of the most important pulse crops in India. It is mainly cultivated during the Kharif season with sowing in June–July and harvested during winter. India is the largest producer and consumer of pigeon pea, and tur dal is a staple protein source in Indian diets, commonly used in dal and sambar. The crop contains about 20–22% protein and plays a crucial role in nutritional security. Besides its dietary value, arhar improves soil fertility through biological nitrogen fixation, is drought tolerant, and provides fodder and fuel wood. It is highly suitable for crop rotation and intercropping systems.

Botanically, pigeon pea is a perennial shrub grown as an annual crop in agriculture. Plants grow 1–3 meters tall and possess a deep taproot system that enables drought tolerance. The crop bears yellow flowers and pods containing 4–6 seeds. It is both self- and cross-pollinated. Its origin is debated, with evidence pointing to India or Africa.

Arhar requires warm climatic conditions. Germination occurs best at 25–30°C, and vegetative growth thrives between 20–35°C. The crop is sensitive to frost, and high temperatures during flowering reduce pod set and yield. It requires about 60–100 cm annual rainfall and tolerates drought well due to its deep roots. However, it is highly sensitive to waterlogging; standing water for more than 24–48 hours can cause root damage and disease.

The crop grows best in well-drained loamy soils with a pH of 6.5–7.5. Slightly alkaline soils are tolerated, but heavy clay and waterlogged soils should be avoided. Maharashtra is the largest producer, followed by Karnataka, Madhya Pradesh, Uttar Pradesh, Telangana, and Gujarat.

Arhar varieties are grouped into early (120–150 days), medium (150–180 days), and late maturity types (180–250 days). Popular varieties include UPAS 120, ICPL 87119 (Asha), Pusa 992, Maruti, and BDN 711. Selection depends on rainfall pattern and cropping system.

Land preparation involves one deep ploughing followed by 2–3 harrowings to create a well-pulverized seedbed with good drainage. Weeds and residues should be removed. Sowing is done during June–July with monsoon onset. The recommended seed rate is 12–15 kg per hectare with spacing of 60–90 cm between rows and 20–30 cm between plants. Line sowing using a seed drill ensures uniform plant population. Seeds should be treated with Carbendazim to prevent fungal infection and inoculated with *Rhizobium* culture to enhance nitrogen fixation. The protective effect of seed treatment generally lasts 30–40 days after germination.

Arhar is mostly rainfed, but irrigation at flowering and pod formation significantly increases yield. Water stagnation must be strictly avoided. Fertilizer recommendation includes 20–25 kg nitrogen, 50–60 kg phosphorus, and 20 kg potassium per hectare. Zinc sulfate should be applied if deficiency symptoms appear. Organic options include 5 tons of farmyard manure per hectare. Weed control is critical between 20–45 days after sowing; hand weeding or intercultivation is recommended. Pendimethalin can be used as a pre-emergence herbicide.

Among diseases, Fusarium wilt caused by *Fusarium udum* is highly destructive, causing 30–60% yield loss. Symptoms include sudden wilting, yellowing of leaves, and plant death. The disease can appear at any stage but is most common between 30–90 days after sowing. Warm soil and continuous pigeon pea cultivation favor infection. Prevention includes using resistant varieties, practicing crop rotation for 3–4 years with cereals, deep summer ploughing, and seed treatment. Soil application of *Trichoderma* at sowing protects roots for approximately 40–50 days. Infected plants should be removed immediately. There is no complete cure once plants wilt; hence prevention before and during early crop growth (first 60 days) is essential.

Sterility Mosaic Disease (SMD) is a viral disease transmitted by mites and can cause 20–90% yield loss. Infected plants show bushy growth, excessive branching, and absence of flowering. Early infection within 30–45 days leads to severe yield reduction. Prevention focuses on resistant varieties and controlling mite vectors. Spray acaricides such as Dicofol at early detection and repeat after 10–15 days if needed. Regular monitoring during the first 60 days after sowing is critical. Removal of infected plants reduces spread.

Phytophthora blight, caused by *Phytophthora drechsleri*, leads to stem rot and wilting, especially during heavy rainfall. The disease spreads rapidly under waterlogged conditions. Proper drainage is the most important preventive measure. Fungicide sprays such as Metalaxyl should be applied at early symptom detection and repeated after 10–12 days during continuous rainy weather. Monitoring is essential during the vegetative stage (30–80 days).

Powdery Mildew, caused by *Erysiphe polygoni*, appears as a white powdery coating on leaves and stems during dry weather with cool nights. Spray sulfur at first appearance and repeat after 10 days if necessary. Disease usually appears after 80 days; hence monitoring should continue until maturity.

Major insect pests include Pod Borer (*Helicoverpa*), Pod Fly, and Aphids. Pod borer is the most destructive, causing 20–50% yield loss by feeding on developing seeds. Install pheromone traps from 40 days after sowing for early detection. Spray Chlorantraniliprole or Spinosad at flowering and repeat after 10–12 days if infestation persists. Neem oil sprays at 7-day intervals are effective in organic farming. Pod fly damages developing seeds and should be controlled through timely insecticide application during pod formation. Aphids suck sap and weaken plants; neem oil or systemic insecticides can be applied at early detection.

Nutrient deficiencies include yellow leaves (nitrogen), poor root growth (phosphorus), leaf margin burn (potassium), and stunted growth (zinc). Balanced fertilization and micronutrient sprays correct symptoms within 10–15 days after application.

Harvesting is done 150–200 days after sowing when pods turn brown and dry. Plants should be cut, dried, and then threshed. Average yield ranges from 1–1.5 tons per hectare, and improved varieties can produce up to 2 tons per hectare under good management.

After harvest, seeds should be dried to about 10% moisture, cleaned, graded, and stored in a dry place. Nutritionally, pigeon pea contains approximately 20–22 g protein, 60 g carbohydrates,

15 g fiber, and 5 mg iron per 100 g. It supports muscle growth, improves digestion, and promotes heart health.

Organic arhar cultivation involves Rhizobium inoculation, neem cake application, *Trichoderma* seed treatment, and biological pest control methods. Integrated Pest Management (IPM) combining resistant varieties, crop rotation, biological control, and need-based chemical sprays significantly reduces yield losses.

Modern technologies include hybrid varieties, integrated nutrient and pest management, mechanized harvesting, and climate-resilient cultivars. Arhar fits well in cropping systems such as intercropping with cotton and soybean or mixed cropping with millets. Regionally, Maharashtra favors long-duration varieties, Karnataka promotes intercropping, Uttar Pradesh grows early-maturing types, and Telangana practices rainfed farming.

Effective disease prevention in arhar requires close monitoring from 30–90 days after sowing, timely spray intervals of 10–15 days during favorable conditions, strict drainage management, and use of resistant varieties. Early detection and preventive practices are key to minimizing yield loss and ensuring stable production.

6. Urad

Urad, commonly known as Black Gram, scientifically named *Vigna mungo*, belongs to the Fabaceae (legume) family. It is one of the most important pulse crops cultivated in India and is mainly grown during the Kharif season (June–July). It is also cultivated in Rabi (September–October) and Summer (February–March) seasons under irrigation. Urad dal is widely used in Indian cuisine for preparing dosa, idli, vada, papad, and various traditional foods. The crop is a rich source of plant protein (22–26%) and contributes significantly to nutritional security.

Botanically, urad originated in India and is an annual herbaceous plant growing 30–100 cm tall. It has a hairy stem, trifoliate leaves, yellow flowers, and pods containing 6–10 black seeds. It is a self-pollinated crop with a deep root system that bears nitrogen-fixing nodules. These nodules fix atmospheric nitrogen, improving soil fertility and reducing fertilizer requirements for subsequent crops. Due to its short duration of 70–100 days, urad fits well in crop rotation and intercropping systems.

Urad grows best at temperatures between 20–35°C, with 25–30°C ideal for germination. It is sensitive to frost and cannot tolerate waterlogging. Excessive rainfall during flowering reduces yield. The crop requires 60–80 cm annual rainfall and performs well under rainfed conditions if drainage is proper. Well-drained loamy soils with a pH of 6.0–7.5 are ideal. Slightly acidic soils are tolerated, but heavy clay and waterlogged soils must be avoided.

Major urad-producing states in India include Madhya Pradesh, Uttar Pradesh, Maharashtra, Andhra Pradesh, Tamil Nadu, Telangana, and Rajasthan. India is the world's largest producer of black gram. Popular varieties include PU 31, T 9, LBG 752, Pant U 19, and TAU 1.

Land preparation involves 1–2 ploughings to obtain fine tilth soil, followed by leveling and removal of weeds and residues. Sowing is done in lines using a seed drill. A seed rate of 15–20 kg per hectare is recommended, with spacing of 30 cm between rows and 10 cm between plants. Seeds should be treated with Carbendazim or Thiram to prevent fungal infections and inoculated with *Rhizobium* to enhance nitrogen fixation. Seed treatment protection generally lasts for about 30–40 days after sowing.

Urad is mostly rainfed, but irrigation during flowering (30–40 days after sowing) and pod formation (45–55 days) improves yield. Light irrigation during dry spells is beneficial, but water stagnation beyond 24 hours must be avoided. Fertilizer recommendations include 15–20 kg nitrogen, 40–50 kg phosphorus, and 20 kg potassium per hectare. Zinc sulfate should be applied if deficiency symptoms appear. Organic farming includes application of 5 tons farmyard manure per hectare.

Weed competition is critical during 15–30 days after sowing. Hand weeding at 20–25 days and intercultivation at 35 days are effective. Pendimethalin may be applied as a pre-emergence herbicide.

Among diseases, Yellow Mosaic Virus (YMV) is the most destructive, causing 30–70% yield loss. It is transmitted by whiteflies. Symptoms include yellow patches, mosaic pattern, stunted growth, and poor pod formation. Infection during the first 40 days causes maximum yield loss. Prevention must begin immediately after germination and continue up to 45 days after sowing. Use resistant varieties, remove infected plants, and spray Imidacloprid at early whitefly detection. Repeat spraying after 10–12 days if whitefly population persists. Organic control includes neem oil spray at 7–10 day intervals and installation of yellow sticky traps. Continuous monitoring every 5–7 days during the first 45 days is essential.

Leaf Spot caused by *Cercospora canescens* produces brown circular spots on leaves and can cause 15–30% yield loss. The disease usually appears 25–35 days after sowing under humid conditions. Spray Mancozeb at first symptom appearance and repeat after 10–14 days if humidity continues. Maintain field sanitation and avoid dense planting to improve air circulation. Monitoring should continue until pod formation stage (60 days).

Powdery Mildew caused by *Erysiphe polygoni* appears as white powdery coating on leaves during dry weather with cool nights, usually after 50–60 days. Spray wettable sulfur at early detection and repeat after 10 days if needed. Neem oil can be used in organic systems.

Wilt caused by *Fusarium oxysporum* results in wilting and plant death. It may appear from 30–70 days after sowing. Prevention includes crop rotation for 3–4 years with non-host crops, seed treatment with fungicide, and soil application of *Trichoderma*. Once severe wilting occurs, plants cannot recover, so prevention during early growth stages (first 50 days) is crucial.

Major pests include Pod Borer, Aphids, Whiteflies, and Thrips. Pod borer damages developing pods and may cause 20–40% yield loss. Install pheromone traps from 25–30 days after sowing for early detection. Spray Chlorantraniliprole or Spinosad at flowering and repeat after 10–12 days if infestation continues. Neem oil sprays at weekly intervals are effective under organic farming. Aphids suck sap and cause leaf curling; neem oil or Imidacloprid can control them. Whiteflies transmit YMV and must be controlled early. Thrips cause leaf curling and flower damage; neem spray or suitable insecticide controls them.

Nutrient deficiencies include pale yellow leaves due to nitrogen deficiency, poor root growth due to phosphorus deficiency, leaf margin burn due to potassium deficiency, and stunted plants due to zinc deficiency. Balanced fertilization and micronutrient foliar spray correct symptoms within 10–14 days.

Harvesting is done 70–100 days after sowing when pods turn black. Harvesting should be timely to prevent shattering losses. Plants are cut and dried before threshing. Average yield ranges from 0.8–1.2 tons per hectare, while improved varieties under good management can yield 1.5–2 tons per hectare.

After harvest, seeds must be dried to about 10% moisture, cleaned, graded, and stored in dry airtight containers to prevent storage pests. Nutritionally, per 100 g of urad contains about 24 g

protein, 60 g carbohydrates, 18 g fiber, 7 mg iron, and 138 mg calcium. It supports muscle strength, improves digestion, prevents anemia, and promotes heart health.

Organic urad cultivation includes Rhizobium inoculation, neem cake application, *Trichoderma* seed treatment, and organic pest control methods. Integrated Pest Management (IPM) combining resistant varieties, crop rotation, biological control agents, pheromone traps, and need-based pesticide application significantly reduces yield losses.

Urad fits well in cropping systems such as rice–urad, wheat–urad, and intercropping with maize or cotton. Regionally, Madhya Pradesh grows large-scale Kharif urad, Uttar Pradesh practices rice-fallow urad, Tamil Nadu grows summer urad under irrigation, and Maharashtra adopts mixed cropping systems.

Effective disease management requires preventive action during the first 40–60 days after sowing, spray intervals of 7–14 days under favorable conditions, strict drainage control, and regular monitoring. Early detection and timely intervention are essential to minimize yield loss and ensure healthy crop production.

7. Masoor

Masoor, commonly known as Lentil, scientifically named *Lens culinaris*, belongs to the Fabaceae (legume) family and is one of the most important Rabi pulse crops in India. It is sown during October–November and harvested in spring after about 100–120 days. Lentil is widely consumed as dal and serves as a rich source of protein (22–25%), making it a vital component of vegetarian diets. The crop improves soil fertility through biological nitrogen fixation and enhances soil structure, making it ideal for crop rotation systems. It is suitable for rainfed farming and performs well in dryland conditions.

Lentil originated in the Near East region and is an annual herbaceous plant growing 30–50 cm tall. It has a weak stem, compound leaves, small white or blue flowers, and pods containing 1–2 seeds. It is a self-pollinated crop with a shallow root system bearing nitrogen-fixing nodules. Due to its shallow roots, it is sensitive to excess soil moisture but well adapted to residual soil moisture conditions.

The crop grows best at temperatures between 15–25°C, with 20–25°C ideal for germination. It tolerates mild frost but is sensitive to high temperatures during flowering, which may reduce pod setting. Lentil requires about 30–40 cm annual rainfall and thrives in dryland areas. However, it is highly sensitive to waterlogging; standing water for more than 24–48 hours may cause root damage and disease.

Lentil grows best in well-drained loamy soils ranging from sandy loam to clay loam with a pH between 6.0 and 7.5. Slightly alkaline soils are tolerated, but heavy and poorly drained soils should be avoided. Major lentil-producing states in India include Madhya Pradesh (largest producer), Uttar Pradesh, Bihar, West Bengal, Rajasthan, and Jharkhand.

Lentil types include small-seeded and large-seeded varieties. Popular improved varieties such as Pusa Vaibhav, PL 406, HUL 57, JL 3, and DPL 62 provide higher yields and improved disease resistance.

Land preparation requires 1–2 ploughings to achieve fine tilth, removal of weeds, and proper drainage arrangements. Sowing is done in October–November using line sowing or seed drill method. The recommended seed rate is 30–40 kg per hectare, with spacing of 25–30 cm between rows and 5–10 cm between plants. Seed treatment with Carbendazim protects against fungal pathogens, and Rhizobium inoculation enhances nitrogen fixation. Seed treatment protection typically lasts 30–40 days after sowing.

Lentil is mostly rainfed, but light irrigation at flowering (60–70 days after sowing) and pod formation (80–90 days) improves yield if soil moisture is low. Excess irrigation must be avoided. Fertilizer recommendations include 15–20 kg nitrogen, 40–50 kg phosphorus, and 20 kg potassium per hectare. Zinc sulfate should be applied if deficiency symptoms appear. Organic farmers may apply 5 tons of farmyard manure per hectare before sowing.

Weed competition is critical during 20–40 days after sowing. Hand weeding or intercultivation during this period ensures healthy crop growth. Pendimethalin may be used as a pre-emergence herbicide.

Among diseases, Fusarium wilt caused by *Fusarium oxysporum* is serious and may cause 30–60% yield loss. Symptoms include yellowing of leaves, wilting, and eventual plant death. The disease can appear from 30 days onward and is favored by warm soil and continuous lentil cultivation. Prevention includes growing resistant varieties, practicing crop rotation for 3–4 years with cereals, deep summer ploughing, and seed treatment. Soil application of *Trichoderma* during sowing provides protection for about 40–50 days. Once plants wilt severely, recovery is unlikely; hence preventive measures during early growth stages (first 50–60 days) are essential.

Rust caused by *Uromyces fabae* produces brown pustules on leaves, usually appearing during flowering and pod stages (70–90 days). It can cause 20–40% yield loss under humid conditions. Spray Mancozeb at early detection and repeat after 10–14 days if humidity persists. Monitoring should continue until maturity.

Ascochyta blight caused by *Ascochyta lentis* produces dark spots on leaves, stems, and pods. It spreads rapidly in cool, wet weather and may cause severe damage. Seed treatment is the first line of defense. Fungicide spray at symptom appearance should be repeated after 10–12 days if favorable conditions continue. Monitoring is important from 40–80 days after sowing.

Powdery Mildew caused by *Erysiphe polygoni* appears as white powdery growth on leaves during dry weather with cool nights, usually after 80 days. Spray sulfur at early symptom appearance and repeat after 10 days if necessary. Neem oil can be used as an organic alternative.

Major insect pests include Pod Borer, Aphids, and Cutworms. Pod borer damages developing seeds and may cause 15–30% yield loss. Install pheromone traps from 40 days after sowing for early detection. Spray recommended insecticides at flowering and repeat after 10–12 days if infestation continues. Aphids suck plant sap and can be controlled with neem oil or Imidacloprid spray. Cutworms damage seedlings and can be controlled by proper soil treatment before sowing.

Nutrient deficiencies include pale leaves due to nitrogen deficiency, stunted growth due to phosphorus deficiency, leaf margin burn due to potassium deficiency, and small leaves due to zinc deficiency. Balanced fertilization and micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done 100–120 days after sowing when plants turn yellow and pods become brown and dry. Plants should be cut and dried before threshing to reduce shattering losses. Average yield ranges from 1–1.5 tons per hectare, while improved varieties can produce up to 2 tons per hectare under good management.

After harvest, seeds must be dried to about 10% moisture, cleaned, graded, and stored in dry conditions to prevent storage pests. Nutritionally, lentils contain about 24 g protein, 60 g

carbohydrates, 11 g fiber, 7 mg iron, and are rich in folate per 100 g. They improve digestion, support heart health, prevent anemia, help manage blood sugar, and provide antioxidants.

Organic lentil cultivation includes *Rhizobium* inoculation, neem cake application, *Trichoderma* seed treatment, and organic pest control measures. Integrated Pest Management (IPM), combining resistant varieties, crop rotation, biological control agents, and need-based chemical sprays, significantly reduces yield losses.

Lentil fits well in cropping systems such as Rice–Lentil and Wheat–Lentil rotations and can be intercropped with mustard. Regionally, Madhya Pradesh practices large-scale Rabi lentil cultivation, Uttar Pradesh adopts mixed cropping, Bihar grows rainfed lentil, and West Bengal cultivates lentil after rice.

Effective disease prevention in lentil requires close monitoring from 30–90 days after sowing, timely fungicide spray intervals of 10–14 days during favorable conditions, proper drainage, resistant varieties, and early removal of infected plants. Preventive management during early crop growth and flowering stages is crucial to minimize yield loss and ensure stable production.

8. Sugarcane

Sugarcane, scientifically known as *Saccharum officinarum*, belongs to the Poaceae (grass) family and is one of the most important commercial cash crops in India. It is the primary source of sugar and plays a vital role in the agricultural economy. India is among the largest producers of sugarcane globally. The crop is grown mainly in tropical and subtropical regions and has a long duration of 10–18 months depending on variety and climate. Besides sugar, sugarcane is used to produce jaggery (gur), khandsari, and ethanol. Its by-products include bagasse (used as fuel and in paper industries), molasses (for alcohol and ethanol production), and press mud (used as fertilizer). The crop supports rural employment through sugar mills and allied industries.

Sugarcane originated in Southeast Asia and is a perennial grass cultivated as an annual crop. It grows 2–4 meters tall and has a thick fibrous stalk rich in sucrose. The plant has long narrow leaves and a fibrous root system. Propagation is done vegetatively using stem cuttings known as setts, usually containing 2–3 buds.

The crop requires a warm climate throughout its growth. Germination occurs best at 25–35°C, vegetative growth at 20–30°C, and ripening at 15–20°C for better sugar accumulation. It is sensitive to frost, which damages stalks and reduces sugar content. Sugarcane requires 75–150 cm annual rainfall and adequate irrigation in low rainfall regions. Bright sunshine and a long growing season are essential for maximum biomass and sugar production.

Sugarcane grows best in deep, fertile, well-drained loamy soils with a pH of 6.5–7.5. It tolerates slightly alkaline soils but cannot withstand prolonged waterlogging. Major sugarcane-producing states in India include Uttar Pradesh (largest producer), Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Gujarat, Bihar, and Haryana.

Varieties are classified as early, mid-late, and late maturing types. Popular varieties include Co 0238, Co 86032, Co 99004, CoC 671, and Co 0118. Hybrid varieties provide higher cane yield and better sugar recovery.

Land preparation involves deep ploughing followed by 2–3 harrowings and proper leveling. Ridges and furrows are formed to facilitate irrigation and drainage. Planting is done during February–March (spring planting) or October–November (autumn planting). Healthy 2–3 budded setts are used at a rate of 6–8 tons per hectare. Row spacing varies from 75–120 cm depending on method. Planting methods include flat, ridge and furrow, and trench methods. Seed treatment with fungicide and hot water treatment (50°C for 30 minutes) prevents seed-borne diseases and provides protection during early growth for about 30–45 days.

Sugarcane requires regular irrigation at critical stages such as germination (0–45 days), tillering (45–90 days), and grand growth stage (90–180 days). Irrigation is typically given every 10–15 days depending on soil and climate. Drip irrigation improves water use efficiency and increases yield.

Fertilizer management is crucial due to high nutrient demand. Recommended doses include 150–250 kg nitrogen, 60–100 kg phosphorus, and 60–120 kg potassium per hectare. Nitrogen should be applied in split doses—basal, tillering, and grand growth stages—to reduce losses and improve uptake. Micronutrients like zinc and iron should be applied if deficiencies appear. Organic options include applying 10–15 tons of farmyard manure per hectare before planting.

Weed control is essential during 30–90 days after planting. Hand weeding, intercultivation, and pre-emergence herbicide such as Atrazine are effective.

Among diseases, Red Rot caused by *Colletotrichum falcatum* is the most serious and may cause 30–70% yield loss. It produces red discoloration with white patches inside the stalk and leads to wilting. Infection often appears from 3–6 months after planting. Prevention includes using resistant varieties, crop rotation for 2–3 years, avoiding ratooning of infected fields, and using disease-free setts. Infected clumps should be uprooted and destroyed immediately. There is no effective cure once the disease is established; therefore, prevention during planting and early growth (first 120 days) is critical.

Smut caused by *Sporisorium scitamineum* produces a black whip-like structure emerging from the growing point and causes stunted growth. It spreads through infected planting material. Hot water treatment of setts before planting prevents infection. Infected plants should be removed as soon as detected. Monitoring is necessary from 2–5 months after planting. Preventive measures must continue throughout vegetative growth.

Wilt caused by *Fusarium* species leads to yellowing and drying of leaves. It appears mainly in mid to late growth stages (4–8 months). Prevention includes crop rotation, proper drainage, and balanced fertilization. Seed treatment and soil application of bioagents like *Trichoderma* reduce incidence for approximately 45–60 days after planting.

Grassy Shoot Disease caused by phytoplasma and transmitted by leafhoppers produces bushy growth and thin leaves. Early infection within 2–4 months severely reduces yield. Prevention includes using disease-free planting material, controlling leafhopper vectors with insecticides, and removing infected plants. Vector control should begin from 30 days after planting and continue at 15-day intervals during active vector presence.

Major pests include Early Shoot Borer, Top Borer, White Grub, and Scale Insects. Early Shoot Borer damages young shoots within the first 60 days after planting. Application of Chlorantraniliprole at early infestation and field monitoring from 20–60 days prevents spread. Top Borer affects the growing point during 90–150 days; pheromone traps help in monitoring and timely insecticide application. White Grubs damage roots and require soil treatment before or at planting. Scale insects suck sap and can be controlled by insecticide spray when infestation appears.

Nutrient deficiencies manifest as yellowing leaves (nitrogen), poor root growth (phosphorus), leaf margin burn (potassium), white streaks (zinc), and yellowing of young leaves (iron). Soil testing and balanced fertilization correct deficiencies within 15–20 days of application.

Harvesting is done 10–18 months after planting when stalks mature and sugar content peaks. Canes are cut close to the ground to maximize yield. Average yield ranges from 60–80 tons per hectare, while improved varieties under good management can produce 100–120 tons per hectare.

After harvest, canes should be transported quickly to sugar mills to avoid sucrose loss due to inversion. The sugar production process includes crushing, juice extraction, clarification, evaporation, crystallization, and centrifugation. By-products such as bagasse, molasses, and press mud are utilized efficiently.

Nutritionally, sugar provides 100% carbohydrates in the form of sucrose and supplies quick energy. However, excessive consumption may cause health problems, while jaggery contains small amounts of minerals.

Organic sugarcane cultivation involves using organic manures, biofertilizers, neem cake, and biological pest control methods. Integrated Pest Management (IPM) combining resistant varieties, crop rotation, biological control, and need-based chemical use reduces yield losses significantly.

Modern technologies include tissue culture seedlings for disease-free planting, drip irrigation, precision farming, mechanized harvesting, and climate-resilient varieties. Sugarcane fits well in cropping systems such as Sugarcane–Wheat rotation, intercropping with pulses, and ratoon cropping systems.

Effective disease prevention requires continuous monitoring from planting up to 8 months of growth. Preventive measures such as seed treatment, vector control, proper drainage, and resistant varieties should be maintained especially during the first 120 days. Fungicide or insecticide sprays, if required, are typically repeated at 10–15 day intervals during favorable conditions. Early detection and strict preventive management significantly reduce yield losses and ensure sustainable sugarcane production.

9. Cotton

Cotton, scientifically known as *Gossypium hirsutum* (the most widely cultivated species), belongs to the Malvaceae family and is one of the most important fiber and cash crops in India and the world. It is popularly called “White Gold” due to its high economic value. India is among the largest cotton producers globally, and the crop is mainly grown during the Kharif season with sowing in June–July and harvesting from October to January. Cotton is the primary source of natural fiber used in the textile and garment industries. It provides employment to millions of farmers and workers in spinning, weaving, and textile manufacturing sectors. Cottonseed oil is used for cooking, and cottonseed cake is an important cattle feed. The crop contributes significantly to export earnings and rural economic stability.

Cotton originated in tropical and subtropical regions and is grown as an annual shrub. Plants grow about 1–2 meters tall with a deep taproot system that allows drought tolerance. The plant produces large yellow or white flowers, which later develop into fruits called bolls. Each boll contains seeds surrounded by cotton fiber. There are four main cultivated species: *Gossypium hirsutum* (American cotton), *Gossypium arboreum* (Desi cotton), *Gossypium barbadense* (Egyptian cotton), and *Gossypium herbaceum*. In India, cotton is grown in three major zones: Northern, Central, and Southern zones. Gujarat is the largest producer, followed by Maharashtra, Telangana, Andhra Pradesh, Punjab, Haryana, Rajasthan, and Madhya Pradesh.

Cotton requires warm climatic conditions. Germination occurs best at 20–30°C, and growth is optimal between 25–35°C. The crop is sensitive to frost and requires a long frost-free period of about 150–180 days. Annual rainfall of 50–100 cm is suitable, but excessive rainfall and waterlogging reduce yield and increase disease incidence. Bright sunshine during flowering and boll development ensures better fiber quality.

The crop grows best in deep black cotton soils (regur soils) and well-drained loamy soils with a pH range of 6.0–8.0. Saline and poorly drained soils should be avoided. Land preparation includes deep ploughing, 2–3 harrowings, leveling, and ensuring proper drainage. Sowing is done during June–July with the onset of monsoon. Seed rate varies between 15–25 kg per hectare depending on variety. Spacing ranges from 60–120 cm between rows and 30–60 cm between plants. Seeds should be treated with fungicides to prevent early fungal diseases and inoculated with biofertilizers for better nutrient uptake. Seed treatment provides protection during the first 30–40 days after sowing.

Irrigation is critical during flowering (60–80 days after sowing) and boll formation (90–120 days). Excess irrigation should be avoided. Drip irrigation improves water efficiency and yield. Fertilizer recommendations include 100–150 kg nitrogen, 50–60 kg phosphorus, and 50–60 kg potassium per hectare. Nitrogen should be applied in split doses—basal, vegetative stage, and flowering stage. Zinc and boron should be applied if deficiency symptoms appear. Organic farmers may apply 10 tons of farmyard manure per hectare.

Weed competition is critical between 20–60 days after sowing. Hand weeding, intercultivation, and pre-emergence herbicide such as Pendimethalin are effective control measures.

Among diseases, Fusarium wilt caused by *Fusarium oxysporum* results in yellowing, wilting, and plant death, leading to 15–30% yield loss. The disease often appears between 30–90 days after sowing. Prevention includes using resistant varieties, crop rotation for 3–4 years with cereals, seed treatment, and maintaining proper drainage. Soil application of bioagents like *Trichoderma* provides protection for about 45 days. Infected plants should be removed immediately, as severely wilted plants cannot recover.

Leaf Curl Virus, transmitted by whiteflies, is one of the most serious viral diseases and can cause 30–70% yield loss. Symptoms include upward leaf curling, thickening, and stunted growth. Infection during the first 50–60 days causes maximum damage. Preventive measures include planting resistant hybrids, controlling whitefly populations with Imidacloprid or other recommended insecticides, and installing yellow sticky traps. Sprays should be applied at early detection and repeated every 10–12 days if whitefly populations remain high. Monitoring should be frequent during the first 60 days.

Alternaria Leaf Spot causes brown circular leaf spots and is common in humid conditions. It usually appears 45–75 days after sowing. Spray Mancozeb at first symptom appearance and repeat after 10–14 days if needed. Maintaining proper plant spacing and field sanitation reduces infection.

Bacterial Blight caused by *Xanthomonas campestris* produces angular leaf spots and can spread rapidly in rainy conditions. Seed treatment and use of resistant varieties are important preventive measures. Copper-based fungicides can be sprayed at symptom appearance and repeated after 10 days if required.

Major pests include Bollworms, Whiteflies, Aphids, and Pink Bollworm. Bollworms damage bolls and cause 20–60% yield loss. Bt cotton hybrids provide resistance to certain bollworms. Pheromone traps should be installed from 40 days after sowing to monitor pest levels. Insecticide application at flowering and boll formation stages, repeated after 10–12 days if infestation persists, is recommended. Pink Bollworm is a major pest in many regions and requires strict integrated pest management including crop residue destruction and timely harvesting. Whiteflies and aphids suck plant sap and can be controlled with neem oil or systemic insecticides applied at early infestation.

Nutrient deficiencies include yellowing leaves (nitrogen), stunted growth (phosphorus), leaf margin burn (potassium), small leaves (zinc), and poor boll development (boron). Balanced fertilization and foliar micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done 150–180 days after sowing when bolls burst open. Cotton is handpicked in multiple stages to ensure quality. In some regions, mechanical harvesting is practiced. Average yield of seed cotton is 1.5–2 tons per hectare, while hybrid and Bt varieties can produce 2.5–3 tons per hectare under good management.

After harvest, cotton is dried and sent for ginning to separate fiber from seeds. Fiber is then spun and woven into textiles. By-products include cottonseed oil and cottonseed cake.

Organic cotton cultivation emphasizes biofertilizers, neem-based pest control, crop rotation, and biological pest management. Integrated Pest Management (IPM) combining resistant varieties, pheromone traps, biological control, and need-based pesticide use significantly reduces yield losses.

Modern technologies in cotton farming include Bt hybrids, drip irrigation, precision farming, mechanized harvesting, and climate-resilient varieties. Cotton fits well in cropping systems such as Cotton–Wheat and Cotton–Soybean rotations and intercropping with pulses. Regionally, Gujarat practices large-scale Bt cotton cultivation, Maharashtra grows rainfed cotton, Punjab and Haryana grow irrigated cotton, and Telangana promotes hybrid cotton.

Effective disease and pest prevention requires close monitoring from 30–120 days after sowing. Preventive sprays, if necessary, are usually repeated at 10–14 day intervals during favorable conditions. Early detection, resistant varieties, balanced nutrition, and proper irrigation management are essential to minimize yield loss and ensure sustainable cotton production.

10. Tea

Tea, scientifically known as *Camellia sinensis*, belongs to the Theaceae family and is one of the most important plantation and beverage crops in the world. India is among the largest producers and consumers of tea globally. Tea is cultivated mainly in tropical and subtropical hilly regions and is a perennial crop with an economic lifespan of about 40–50 years. The tender shoots consisting of two leaves and a bud are harvested and processed into black tea, green tea, oolong tea, and white tea depending on the processing method. Tea is a major export commodity, a source of foreign exchange, and provides employment to millions of plantation workers. It is valued for its antioxidants and contributes significantly to rural economies.

Tea originated in China and is an evergreen perennial shrub maintained at a height of 1–1.5 meters through regular pruning to facilitate plucking. It has dark green glossy leaves, small white fragrant flowers, and a deep root system that allows efficient nutrient uptake. Two main botanical varieties are grown: *Camellia sinensis* var. *assamica* (Assam type) and *Camellia sinensis* var. *sinensis* (China type), along with improved hybrid clones that provide higher yield and better quality.

Tea thrives in warm and humid conditions. The ideal temperature range is 20–30°C. It is sensitive to frost and extreme heat. High humidity and well-distributed annual rainfall of 150–300 cm are essential. Tea grows best at altitudes between 600–2000 meters. Deep, well-drained acidic soils rich in organic matter with a pH of 4.5–5.5 are ideal. Good moisture retention without waterlogging is crucial.

Major tea-growing regions in India include Assam (largest producer), West Bengal (Darjeeling), Tamil Nadu (Nilgiris), Kerala, and Karnataka. Assam produces strong black tea, Darjeeling is known for aromatic high-quality tea, and the Nilgiris region produces tea almost year-round.

Tea is propagated either through seeds or vegetative cuttings in nurseries. Clonal propagation ensures uniform quality. Planting is usually done during June–July with spacing of 1–1.5 meters between plants. Shade trees are planted to protect bushes from direct sunlight and maintain microclimate. Regular irrigation is provided during dry seasons, often through sprinkler systems. Waterlogging must be strictly avoided as it leads to root diseases.

Tea requires high nutrient input due to continuous plucking. Recommended fertilizer doses include 200–300 kg nitrogen, 40–60 kg phosphorus, and 120–150 kg potassium per hectare annually, applied in frequent split doses throughout the growing season. Organic options include compost and vermicompost. Soil testing ensures balanced fertilization.

Pruning and training are essential cultural operations in tea cultivation. Light, medium, and rejuvenation pruning are practiced to maintain bush height, stimulate new shoot growth, and remove diseased wood. Plucking is done every 7–15 days depending on growth rate, and skilled labor is required to harvest “two leaves and a bud” for quality production.

Among diseases, Blister Blight caused by *Exobasidium vexans* is one of the most serious fungal diseases, causing 20–40% yield loss. It produces blister-like lesions on young leaves and reduces shoot quality. The disease thrives in cool, humid conditions, especially during monsoon. Prevention involves maintaining proper spacing, pruning for aeration, and regular monitoring during rainy seasons. Copper-based fungicide sprays should be applied at the first appearance of symptoms and repeated at 10–14 day intervals during favorable conditions. Monitoring is critical during flush periods when new shoots appear, typically every 7–10 days in high humidity.

Root Rot diseases occur in poorly drained soils and cause wilting and root decay. Prevention focuses on proper drainage, avoiding water stagnation, and maintaining soil health. Soil drenching with recommended fungicides or application of biological agents such as *Trichoderma* provides protection for about 30–45 days. Monitoring should be continuous in areas prone to heavy rainfall.

Red Rust is an algal disease causing reddish patches on leaves, reducing photosynthetic efficiency. It is often associated with nutrient imbalance and poor bush vigor. Balanced fertilization and improved field sanitation help prevent occurrence. If needed, copper sprays may be applied and repeated after 10–15 days.

Major pests include Tea Mosquito Bug, Red Spider Mite, and Tea Looper. Tea Mosquito Bug sucks sap from tender shoots, causing 15–30% yield loss. It is more active during dry months. Regular monitoring every 7 days during peak season is necessary. Insecticide sprays should be applied at early infestation and repeated after 10–14 days if required. Red Spider Mite causes leaf discoloration under hot and dry conditions. Miticide application at early detection and maintenance of field humidity reduce damage. Tea Looper feeds on leaves and can be controlled through biological control agents or need-based insecticide sprays.

Nutrient deficiencies in tea manifest as pale leaves (nitrogen), poor root growth (phosphorus), leaf edge burn (potassium), and interveinal yellowing (magnesium). Soil testing and balanced fertilization correct deficiencies within 15–20 days of application.

Harvesting and processing determine tea quality. For black tea production, steps include withering, rolling, fermentation (oxidation), drying, and grading. Green tea processing skips fermentation. Proper drying, grading, and packing are essential to maintain quality. Made tea yield averages 1500–2500 kg per hectare depending on management and variety.

Tea contains caffeine, polyphenols, antioxidants, and theanine. It improves alertness, supports heart health, boosts immunity, and may reduce stress. Organic tea cultivation involves organic compost, biofertilizers, neem-based pest control, and avoidance of synthetic chemicals.

Yield losses due to blister blight, mosquito bug, and red spider mite can range from 10–40%, but integrated pest and disease management significantly reduces losses. Modern technologies such as clonal varieties, precision nutrient management, integrated pest management (IPM), mechanized plucking, and climate-resilient cultivars improve productivity and sustainability.

Effective disease and pest prevention in tea requires continuous year-round monitoring due to its perennial nature. Preventive fungicide or insecticide applications, when necessary, are typically repeated at 10–14 day intervals during high-risk seasons. Good drainage, balanced nutrition, proper pruning, and use of resistant clones are essential to maintain healthy bushes for decades of productive life.

11. Wheat

Wheat, scientifically known as *Triticum aestivum*, belongs to the Poaceae (grass) family and is one of the most important staple cereal crops in the world. In India, it is the second most cultivated food grain after rice and is primarily grown during the Rabi season from October to April. Wheat forms a major part of daily diets in the form of chapati, bread, biscuits, pasta, and other products. It provides carbohydrates, plant protein, dietary fiber, B-vitamins, iron, and essential minerals. Wheat is vital for national food security and supports millions of farmers, flour mills, bakeries, breweries, and feed industries. Wheat bran is used as livestock feed, and straw is used as fodder.

Wheat requires cool climatic conditions during growth and warm, dry weather during maturity. Germination occurs best at 20–25°C. The tillering stage prefers 16–20°C, and grain filling requires 14–18°C. High temperatures during grain filling reduce yield and grain quality. The crop requires 50–100 cm annual rainfall and supplemental irrigation in low rainfall regions. It is sensitive to waterlogging, which increases fungal disease incidence. Wheat grows best in well-drained loam to clay loam soils with a pH range of 6.0–7.5. Fertile soils rich in organic matter are ideal, while saline and highly acidic soils should be avoided.

Major wheat-producing states in India include Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan, Bihar, Gujarat, and Maharashtra. Punjab and Haryana are known as the “Wheat Bowl of India.” Popular varieties include HD 2967, PBW 343, WH 1105, DBW 187, Lok 1, and HI 1544 (Pusa Wheat), which are known for high yield and disease resistance.

Land preparation involves 2–3 ploughings, proper leveling, and removal of weeds and residues. Sowing is done during October–November using broadcasting, drill sowing, or zero tillage methods (widely practiced in Punjab and Haryana). The recommended seed rate is 100–125 kg per hectare.

Irrigation is crucial at specific stages: Crown Root Initiation (CRI) at 20–25 days after sowing, tillering, booting, flowering, and grain filling. CRI is the most critical stage. Over-irrigation causes lodging and increases disease risk. Fertilizer recommendations generally include 100–150 kg nitrogen, 50–60 kg phosphorus, and 40–50 kg potassium per hectare. Nitrogen is applied in split doses—half at sowing and half at CRI stage. Zinc sulfate should be applied if zinc deficiency appears.

Among diseases, Rust (stem, leaf, and stripe rust) caused by *Puccinia* species is one of the most serious and may cause 20–50% yield loss. Symptoms include orange or brown pustules on leaves and stems, premature drying, and poor grain filling. It thrives in cool, humid weather. Prevention includes growing resistant varieties, early sowing, balanced fertilization, and avoiding excessive nitrogen. Monitoring should begin from 45 days after sowing and continue until grain filling. At first symptom detection, spray Propiconazole or Tebuconazole and repeat after 10–14 days if favorable conditions persist. Organic options include neem-based biofungicides applied at similar intervals.

Powdery Mildew caused by *Blumeria graminis* appears as white powdery growth on leaves during cool, humid weather, typically from 50–70 days after sowing. It can cause 15–30% yield loss. Spray sulfur fungicide at early symptom appearance and repeat after 10 days if necessary. Diluted cow milk spray may reduce infection in organic systems. Monitoring during tillering and booting stages is essential.

Karnal Bunt caused by *Tilletia indica* infects grains, producing black powder and foul odor, causing 10–20% loss and affecting export quality. It is favored by humid conditions during flowering. Prevention includes using certified seeds, crop rotation, and Carbendazim seed treatment before sowing. Field monitoring during flowering (70–90 days) is crucial. There is no effective curative spray after infection; preventive seed treatment and resistant varieties are key.

Loose Smut caused by *Ustilago tritici* replaces grains with black powdery masses. It spreads through infected seeds. Seed treatment with Carboxin before sowing completely prevents disease. Protection lasts throughout the crop season when treated properly.

Septoria Leaf Blotch caused by *Zymoseptoria tritici* produces brown irregular spots and reduces photosynthesis. It occurs in cool, wet conditions during 50–90 days after sowing. Spray Mancozeb at early symptom appearance and repeat after 10–14 days if necessary.

Major pests include Aphids, Termites, and Armyworms. Aphids suck sap and may cause up to 35% yield loss if uncontrolled. Monitor from 40 days onward and spray Imidacloprid at early detection; repeat after 10 days if infestation persists. Neem oil can be used in organic farming. Termites damage roots and are common in dry soils; soil treatment with Chlorpyrifos before sowing or neem cake application prevents infestation. Armyworms feed on leaves; pheromone traps and timely application of Chlorantraniliprole control outbreaks.

Nutrient deficiencies include yellowing of older leaves (nitrogen), stunted growth (phosphorus), brown leaf edges (potassium), white streaks (zinc), and yellowing of young leaves (iron). Balanced fertilization and micronutrient sprays correct symptoms within 10–15 days.

Harvesting occurs in March–April when grains become hard and golden. Harvesting may be manual or by combine harvester. Average yields range from 3–5 tons per hectare, while high-yielding varieties under good management can produce 6–7 tons per hectare.

Preventive farming practices include using certified seeds, crop rotation (e.g., Wheat–Pulses), maintaining proper spacing, balanced fertilizer use, timely irrigation, field sanitation, and regular monitoring every 7–10 days from 30 days after sowing until maturity. Organic wheat cultivation involves seed treatment with *Trichoderma*, application of farmyard manure or vermicompost, neem cake soil treatment, Jeevamrutham and Panchagavya sprays, and biofertilizers such as Azotobacter and PSB.

Yield losses from rust (20–50%), powdery mildew (15–30%), Karnal bunt (10–20%), and aphids (up to 35%) can be minimized through early monitoring and timely intervention. Preventive fungicide sprays, if required, are usually repeated at 10–14 day intervals during favorable

weather conditions. Continuous field monitoring from 30–100 days after sowing is essential to detect early infection and prevent spread.

Modern technologies in wheat farming include zero tillage, laser land leveling, drip irrigation, precision agriculture, drone-based spraying, and climate-resilient varieties. Regionally, Punjab and Haryana adopt zero tillage and combine harvesting; Madhya Pradesh focuses on high-protein wheat; Uttar Pradesh practices irrigated wheat farming; and Rajasthan promotes drought-tolerant varieties.

Effective wheat disease management relies on resistant varieties, timely sowing, balanced nutrition, and careful irrigation management. Early detection within 7–10 days of symptom appearance and immediate control measures significantly reduce yield losses and ensure stable wheat production.

12. Maize

Maize, scientifically known as *Zea mays*, belongs to the Poaceae (grass) family and is one of the most versatile cereal crops in the world. It is often called the “Queen of Cereals” because of its high yield potential and adaptability. In India, maize is mainly grown during the Kharif season (June–July) and also cultivated in Rabi and Summer seasons under irrigation. It serves multiple purposes: human food, poultry and cattle feed, fodder, and industrial raw material for starch, oil, glucose, ethanol, cornflakes, and biofuel production. A major portion of maize production is used in the poultry feed industry, and the crop contributes significantly to rural employment and export earnings.

Maize requires warm climatic conditions. Germination occurs best at 18–25°C, and vegetative growth thrives between 21–32°C. It is highly sensitive to frost and low temperatures. High temperature stress during flowering, especially during tasseling and silking stages, reduces pollination and kernel formation. The crop requires 60–120 cm annual rainfall with good distribution. It is sensitive to waterlogging; standing water for more than 24–48 hours can damage roots and increase disease incidence. Well-drained loamy soils rich in organic matter with a pH of 5.5–7.5 are ideal. Saline and poorly drained soils should be avoided.

Major maize-producing states in India include Karnataka (leading producer), Madhya Pradesh, Maharashtra, Telangana, Andhra Pradesh, Bihar, Uttar Pradesh, and Rajasthan. Popular varieties and hybrids include DHM 117, HQPM 1 (Quality Protein Maize), Vivek Hybrid 9, Ganga 11, Pusa HM-4, and Bio 9681. Types of maize include flint corn, dent corn, sweet corn, popcorn, and waxy corn.

Land preparation involves 2–3 ploughings to achieve fine tilth, followed by leveling for proper drainage. Sowing is done during June–July (Kharif), October–November (Rabi), and January–February (Summer). The recommended seed rate for hybrids is 18–25 kg per hectare with spacing of 60–75 cm between rows and 20–25 cm between plants. Dibbling, seed drill, and ridge-and-furrow methods are commonly used.

Irrigation is critical at knee-high stage (25–30 days after sowing), tasseling (45–55 days), silking (55–65 days), and grain filling (70–90 days). Water stress during silking can reduce yield drastically. Excess irrigation should be avoided. Fertilizer recommendations include 120–150 kg nitrogen, 60 kg phosphorus, and 40 kg potassium per hectare. Nitrogen should be applied in split doses: basal at sowing, at knee-high stage, and at tasseling. Zinc sulfate should be applied if deficiency symptoms appear.

Among diseases, Turcicum Leaf Blight (TLB) caused by *Exserohilum turcicum* produces long cigar-shaped gray lesions on leaves and can cause 20–40% yield loss. It appears usually 30–50 days after sowing under cool and humid conditions. Prevention includes resistant hybrids, crop rotation, balanced nitrogen application, and proper spacing for air circulation. Spray Mancozeb or Propiconazole at early symptom detection and repeat after 10–14 days if humidity persists.

Organic soil application of *Trichoderma* at sowing provides protection for approximately 30–40 days.

Maydis Leaf Blight caused by *Bipolaris maydis* produces small brown rectangular spots and leaf drying, often during 35–60 days after sowing. Resistant hybrids and timely Carbendazim spray at symptom appearance, repeated after 10–12 days if needed, effectively control the disease.

Downy Mildew caused by *Peronosclerospora sorghi* produces yellow stripes, stunted growth, and poor cob formation. It can cause 30–60% yield loss if infection occurs early (within first 30 days). Seed treatment with Metalaxyl before sowing provides early protection for about 30 days. Infected plants should be removed immediately to prevent spread. Monitoring is critical from emergence to 40 days after sowing.

Common Rust caused by *Puccinia sorghi* produces brown pustules on leaves, usually appearing after 40–60 days. It thrives in moderate temperatures and humidity. Spray Tebuconazole at early detection and repeat after 10–14 days if required.

Banded Leaf and Sheath Blight caused by *Rhizoctonia solani* produces white bands on leaves and sheath, commonly during high humidity after 50 days. Proper drainage and balanced fertilization reduce severity. Spray Validamycin at symptom appearance and repeat after 10–12 days if favorable conditions persist.

Major pests include Fall Armyworm, Stem Borer, and Shoot Fly. Fall Armyworm can cause up to 70% yield loss and produces windowpane feeding damage on leaves. Monitoring should begin 10–15 days after emergence. Install pheromone traps for early detection. Spray Chlorantraniliprole at early infestation and repeat after 7–10 days if larvae persist. Neem oil sprays at 5–7 day intervals can reduce early infestation in organic systems.

Stem Borer causes “dead heart” symptoms during 20–40 days after sowing. Apply Carbofuran granules at early stage or release *Trichogramma* parasitoids as biological control. Shoot Fly damages young seedlings within first 15–25 days. Early sowing and Imidacloprid seed treatment prevent early damage.

Nutrient deficiencies include yellowing of lower leaves (nitrogen), purple leaves (phosphorus), leaf edge scorching (potassium), white banding (zinc), and interveinal yellowing (magnesium). Soil testing and balanced fertilization correct symptoms within 10–15 days.

Harvesting is done when husk turns brown and grains become hard with moisture content of 20–25%. Average yield ranges from 4–6 tons per hectare, while hybrids can produce 7–10 tons per hectare under good management. After harvest, grains should be dried to 12–14% moisture and stored in ventilated dry conditions to prevent insect and rodent damage. Metal bins or Pusa bins are recommended for storage.

Organic maize cultivation includes seed treatment with *Trichoderma*, farmyard manure application, biofertilizers like Azotobacter and PSB, neem cake soil treatment, and Panchagavya foliar spray.

Yield losses due to leaf blights (20–40%), downy mildew (30–60%), fall armyworm (up to 70%), and stem borer (20–30%) can be minimized through integrated management. Continuous monitoring from 10–90 days after sowing is essential. Preventive fungicide or insecticide sprays, when needed, are generally repeated at 7–14 day intervals depending on pest pressure and weather conditions.

Modern maize farming uses hybrid seed technology, precision nutrient management, drip irrigation, drone spraying, and climate-smart varieties. Regionally, Karnataka focuses on hybrid maize for poultry feed, Telangana and Andhra Pradesh grow Rabi maize under irrigation, Bihar promotes Quality Protein Maize, and Madhya Pradesh practices large-scale commercial cultivation.

Effective disease prevention requires resistant hybrids, seed treatment, crop rotation, proper drainage, balanced nutrition, and regular monitoring. Early intervention within 7–10 days of symptom appearance significantly reduces disease spread and protects yield.

13. Jowar

Jowar, commonly known as sorghum, scientifically named *Sorghum bicolor*, belongs to the Poaceae (grass) family and is one of the major millet crops grown in India. It is highly valued for its drought tolerance and adaptability to semi-arid climates. Jowar is cultivated in both Kharif (June–July) and Rabi (September–October) seasons and serves as food, fodder, and industrial raw material. It plays a crucial role in food security in dry regions where rainfall is limited. The grain is rich in carbohydrates, protein, iron, and dietary fiber, and it is gluten-free, making it suitable for people with gluten intolerance. It is used to prepare rotis, porridge, and snacks, while the fodder is essential for livestock. It is also used in ethanol production, starch manufacturing, brewing, and animal feed industries.

Jowar grows well in warm climates. Germination occurs best at 25–30°C, and optimal growth takes place between 26–34°C. It tolerates high heat and drought conditions but is sensitive to frost. The crop requires 40–100 cm annual rainfall and performs well under rainfed conditions. It can grow in a wide range of soils, but loamy to clay loam soils with good drainage are ideal. The pH range of 6.0–8.5 is suitable, and it can tolerate moderately saline soils.

Maharashtra is the largest producer of jowar in India, followed by Karnataka, Telangana, Andhra Pradesh, Madhya Pradesh, Rajasthan, and Tamil Nadu. Jowar types include grain sorghum, fodder sorghum, and sweet sorghum (for ethanol). Popular varieties include CSV 15, CSV 20, CSH 14, CSH 16, and M 35-1 (a well-known Rabi variety).

Land preparation involves deep ploughing followed by 2–3 harrowings to obtain fine tilth. Proper leveling ensures good drainage. Sowing is done in Kharif during June–July and in Rabi during September–October. The seed rate is 8–10 kg per hectare for grain types and 25–30 kg per hectare for fodder types. Spacing of 45 cm between rows and 12–15 cm between plants is recommended. Line sowing using a seed drill ensures uniform crop stand.

Jowar is mostly rainfed, but irrigation at critical stages such as booting, flowering, and grain filling significantly improves yield. Waterlogging should be avoided as it promotes disease. Fertilizer recommendations include 80–100 kg nitrogen, 40 kg phosphorus, and 40 kg potassium per hectare. Nitrogen is applied in two splits: basal and at 30 days after sowing. Zinc sulfate may be applied if deficiency symptoms appear.

Among diseases, Anthracnose caused by *Colletotrichum graminicola* is common and may cause 20–40% yield loss. It produces red or brown leaf spots, stem lesions, and grain discoloration. It often appears from 30–60 days after sowing under humid conditions. Prevention includes resistant varieties, crop rotation, and seed treatment with Thiram. Spray Mancozeb at first symptom detection and repeat after 10–14 days if humidity persists. Seed treatment provides early protection for about 30 days after sowing.

Downy Mildew caused by *Peronosclerospora sorghi* results in yellow stripes, stunted growth, and poor ear formation, causing 30–50% yield loss if infection occurs within the first 30–40

days. Metalaxyl seed treatment before sowing provides protection for about 25–30 days. Infected plants should be removed immediately. Monitoring is essential from seedling stage up to 45 days after sowing.

Grain Mold, caused by fungi such as *Fusarium* and *Aspergillus*, affects grain quality especially under high humidity during maturity. It can cause 10–30% yield and quality loss. Prevention includes timely harvesting when grains are mature and proper drying to moisture below 13%. Avoid delayed harvesting beyond 10–15 days after physiological maturity.

Smut caused by *Sporisorium sorghi* produces black powdery masses in ear heads. It spreads through infected seeds. Seed treatment with Carboxin before sowing effectively prevents disease throughout the season.

Rust caused by *Puccinia purpurea* produces brown pustules on leaves and reduces photosynthesis. It appears usually after 40–60 days under humid conditions. Spray Propiconazole at first appearance and repeat after 10–12 days if needed.

Major pests include Shoot Fly, Stem Borer, Aphids, and Midge. Shoot Fly causes “dead heart” symptoms in seedlings within 10–25 days after sowing. Early sowing and Imidacloprid seed treatment provide protection during early stages. Stem Borer damages stems and may require Carbofuran application at early detection. Aphids suck sap and can be controlled by Dimethoate spray or neem oil in organic systems. Midge damages developing grains during flowering; early sowing and Chlorpyrifos spray at flowering reduce infestation.

Nutrient deficiencies include yellowing of older leaves (nitrogen), stunted growth (phosphorus), leaf edge scorching (potassium), white bands (zinc), and yellowing of young leaves (iron). Balanced fertilization and micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done when grains become hard and panicles turn brown. Delayed harvesting increases risk of grain mold. Average yield ranges from 2–3 tons per hectare, while hybrids can produce 4–5 tons per hectare. After harvest, grains should be sun-dried to below 13% moisture and stored in dry containers to prevent storage pests.

Organic jowar cultivation includes farmyard manure application, seed treatment with *Trichoderma*, neem cake soil incorporation, Panchagavya foliar spray, and biofertilizers like Azospirillum. Integrated Pest Management (IPM) combining resistant varieties, early sowing, crop rotation, and need-based pesticide application reduces yield losses significantly.

Modern technologies in jowar farming include hybrid seed technology, drought-tolerant varieties, precision nutrient management, drip irrigation, mechanized harvesting, and climate-smart agriculture practices. Regionally, Maharashtra is known for Rabi jowar cultivation, Karnataka promotes dual-purpose varieties, Telangana practices rainfed farming, and Tamil Nadu grows fodder sorghum.

Effective disease prevention in jowar requires monitoring from 15–60 days after sowing for early diseases and during flowering to maturity for grain mold and rust. Preventive fungicide or

insecticide sprays, if necessary, are generally repeated at 10–14 day intervals during favorable conditions. Early detection within 7–10 days of symptom appearance, use of resistant varieties, seed treatment, balanced nutrition, and proper drainage are essential to minimize yield losses and maintain stable production.

14. Bajra

Bajra, commonly known as Pearl Millet and scientifically called *Pennisetum glaucum* (syn. *Cenchrus americanus*), belongs to the Poaceae family. It is one of the most important millet crops grown in arid and semi-arid regions of India. Bajra is highly drought-tolerant and capable of growing in sandy and poor soils where other cereals fail. It is mainly cultivated during the Kharif season (June–September) and plays a major role in ensuring food and fodder security in dryland agriculture.

Bajra is a staple food in Rajasthan, Gujarat, Haryana, and parts of Uttar Pradesh. It contains about 10–12% protein and is rich in fiber, iron, and calcium. Being gluten-free and having a low glycemic index, it is suitable for people with diabetes and heart-related conditions. It is used for making rotis, porridge, khichdi, snacks, and livestock feed. Due to its high heat and drought tolerance, it is considered a climate-resilient crop.

The crop originated in Africa and was introduced to India thousands of years ago. It has a deep root system extending up to 2 meters, which helps it survive prolonged dry spells. Plants have thick stems and compact ear heads (spikes). Bajra is both self- and cross-pollinated and has a short duration of 75–100 days.

Bajra thrives in warm climates. Germination occurs best at 25–30°C, and growth is optimal at 30–35°C. It can tolerate temperatures up to 40°C but is sensitive to frost. The crop requires 40–75 cm annual rainfall and performs well under low rainfall conditions. It needs full sunlight and some varieties are photoperiod-sensitive.

It grows best in sandy loam to light textured soils with good drainage. The ideal soil pH ranges from 6.5–8.5. Bajra tolerates saline and alkaline soils but cannot withstand waterlogging. Rajasthan is the largest producer, contributing more than 40% of India's production, followed by Uttar Pradesh, Haryana, Gujarat, Maharashtra, and Madhya Pradesh.

Improved varieties include HHB 67 Improved, RHB 177, ICTP 8203, Pusa Composite 61, and GHB 558. Hybrid varieties such as HHB 223 and RHB 121 provide higher yields and better disease resistance.

Land preparation requires one deep ploughing followed by 1–2 harrowings to create fine tilth. Proper leveling ensures uniform sowing and drainage. Sowing is done during June–July with the onset of monsoon. The recommended seed rate is 3–5 kg per hectare with spacing of 45 cm between rows and 10–15 cm between plants. Line sowing using a seed drill is preferred. Seed treatment with Thiram or Metalaxyl before sowing protects seedlings from early fungal infections for about 20–30 days after emergence.

Bajra is mostly rainfed, but irrigation at tillering (20–25 days), flowering (40–50 days), and grain filling (55–70 days) increases yield. Over-irrigation should be avoided to prevent root diseases. Fertilizer recommendation includes 60–80 kg nitrogen, 30–40 kg phosphorus, and 20–30 kg

potassium per hectare. Nitrogen should be split into two doses—half at sowing and half at tillering. Zinc sulfate (25 kg/ha) may be applied if deficiency appears. Organic cultivation can include 5–10 tons of farmyard manure per hectare.

The critical weed competition period is 20–40 days after sowing. Hand weeding, intercultivation, or Atrazine herbicide can be used for control.

Among diseases, Downy Mildew caused by *Sclerospora graminicola* is the most serious and can cause 20–40% yield loss. It produces yellow streaks on leaves, white downy growth on the underside, and “green ear” formation. It usually appears within 15–30 days after sowing under humid conditions. Preventive measures include using resistant hybrids, Metalaxyl seed treatment, and removal of infected plants immediately. Monitoring should begin from 10 days after emergence until 40 days. If infection appears, removal of affected plants within 5–7 days reduces spread.

Ergot caused by *Claviceps fusiformis* produces honey-like secretion from florets and dark sclerotia replacing grains, leading to 10–30% yield loss. It occurs during flowering under humid weather. Prevention includes using clean certified seeds, crop rotation, and Carbendazim spray at early flowering stage. Sprays may be repeated after 10–12 days if rainfall persists. Avoid late sowing to reduce disease incidence.

Smut caused by *Tolyposporium penicillariae* produces black powdery masses in grains. It is seed-borne and can be prevented effectively by seed treatment with Thiram before sowing. Properly treated seeds remain protected for the entire crop season.

Rust caused by *Puccinia substriata* produces brown pustules on leaves, typically appearing 40–60 days after sowing in humid conditions. Spray Mancozeb or Propiconazole at first symptom appearance and repeat after 10–14 days if favorable conditions continue.

Major pests include Shoot Fly, Stem Borer, and Grasshoppers. Shoot Fly causes dead heart symptoms within 10–20 days after sowing. Early sowing and Imidacloprid seed treatment provide early protection. Stem Borer damages stems and may require Carbofuran application at early infestation. Grasshoppers feed on leaves and can be controlled with Chlorpyrifos spray or neem oil in organic systems.

Nutrient deficiencies appear as pale yellow leaves (nitrogen), poor root growth (phosphorus), leaf margin burn (potassium), and white streaks (zinc). Soil testing and micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done 75–100 days after sowing when ear heads turn brown and grains are hard. Average grain yield ranges from 1.5–2.5 tons per hectare, while hybrids can produce 3–4 tons per hectare. Fodder yield ranges from 8–10 tons per hectare. After harvest, grains should be sun-dried to below 12% moisture and stored in airtight containers to prevent storage pests.

Organic bajra cultivation includes seed treatment with *Trichoderma*, farmyard manure application, neem cake incorporation, Panchagavya sprays, and biofertilizers. Integrated Pest

and Disease Management (IPM) combining resistant varieties, early sowing, seed treatment, crop rotation, and need-based fungicide or insecticide sprays significantly reduces yield losses.

Yield losses from downy mildew (20–40%), ergot (10–30%), shoot fly (15–25%), and stem borer (around 20%) can be minimized through early detection. Monitoring should be continuous from 10–50 days after sowing for major diseases and pests. Preventive sprays, if required, are generally repeated at 10–14 day intervals under favorable weather conditions.

Modern technologies in bajra farming include hybrid seed technology, climate-resilient varieties, integrated nutrient and pest management, drip irrigation in drylands, and mechanized sowing. Regionally, Rajasthan practices rainfed hybrid cultivation, Haryana adopts mechanized sowing, Gujarat promotes dual-purpose varieties, and Maharashtra focuses on drought-resistant farming.

Effective disease prevention in bajra depends on resistant varieties, proper seed treatment, early sowing, balanced nutrition, and field monitoring every 7–10 days during early growth and flowering stages. Timely intervention within one week of symptom appearance prevents spread and ensures stable production in dryland ecosystems.

15. Ragi

Ragi, also known as Finger Millet and scientifically called *Eleusine coracana*, belongs to the Poaceae (grass) family and is an important nutri-cereal grown widely in India and Africa. It is highly valued for its exceptional nutritional quality, especially its high calcium and iron content. Ragi is mainly cultivated during the Kharif season (June–July), and in some regions it is also grown during Rabi under irrigation. The crop is drought-tolerant, suitable for dryland farming, and plays an important role in nutritional security.

Ragi is a staple food in Karnataka, Tamil Nadu, Andhra Pradesh, and Telangana. It contains about 300–350 mg calcium per 100 g, making it one of the richest plant sources of calcium. It is also a good source of iron, fiber, and protein, and it is gluten-free with a low glycemic index, making it suitable for diabetic patients. It is consumed in the form of porridge, roti, dosa, malt, and baby food preparations.

The crop originated in East Africa and was introduced to India centuries ago. It is a small-seeded cereal characterized by finger-like ear heads. The plant has a deep root system that enables it to withstand drought conditions. It is a short-duration crop, maturing in 90–120 days.

Ragi grows best in warm climates. Germination occurs at 20–30°C, and growth is optimal between 25–35°C. It tolerates moderate drought but is sensitive to severe frost. It requires 50–100 cm annual rainfall and performs well under rainfed conditions. It can grow up to 2300 meters above sea level, making it suitable for hill regions.

The crop prefers well-drained loamy or red soils with a pH range of 5.0–7.5. It tolerates poor soils but does not tolerate waterlogging. Karnataka is the largest producer, contributing more than 50% of India's ragi production, followed by Tamil Nadu, Andhra Pradesh, Telangana, Uttarakhand, Odisha, and Maharashtra.

Popular varieties include GPU 28, GPU 67, MR 1, PR 202, and Indaf 9. Hybrid varieties such as KMR 204 and KMR 301 provide higher yield potential and disease resistance.

Land preparation requires 2–3 ploughings to achieve fine tilth and proper leveling. Sowing is done during June–July in Kharif and October in irrigated Rabi areas. The seed rate is 8–10 kg per hectare with spacing of 30 cm between rows and 10 cm between plants. Sowing methods include broadcasting, line sowing, and transplanting of nursery-raised seedlings (widely practiced in Karnataka). Seed treatment with Thiram or Carbendazim before sowing protects seedlings from early fungal infection for about 20–30 days after emergence.

Ragi is mostly rainfed, but irrigation during critical stages—tillering (20–25 days), flowering (45–55 days), and grain filling (60–75 days)—improves yield. Fertilizer recommendations include 40–60 kg nitrogen, 20–30 kg phosphorus, and 20 kg potassium per hectare. Nitrogen

should be applied in two splits: basal and at tillering stage. Organic farmers may apply 5 tons of farmyard manure per hectare.

Weed competition is severe during 20–40 days after sowing. Hand weeding, intercultivation, or pre-emergence herbicide such as Isoproturon helps control weeds effectively.

Among diseases, Blast caused by *Pyricularia grisea* is the most serious and may cause 30–50% yield loss. It produces leaf spots, neck rot, and poor grain filling. It usually appears from 25–60 days after sowing, especially under humid and cloudy conditions. Prevention includes growing resistant varieties, balanced nitrogen application (avoid excess nitrogen), proper spacing, and seed treatment. At first symptom appearance, spray Carbendazim or Tricyclazole and repeat after 10–14 days if humid conditions persist. Monitoring should begin from 20 days after sowing until grain filling stage. Early spraying within 5–7 days of symptom detection significantly reduces spread.

Brown Spot caused by *Drechslera nodulosa* produces brown circular spots on leaves and may cause 15–25% yield loss. It usually appears during later vegetative growth (40–70 days). Spray Mancozeb at early symptom appearance and repeat after 10–12 days if necessary. Maintaining proper plant nutrition and spacing reduces incidence.

Downy Mildew caused by *Sclerophthora macrospora* results in yellowing, stunted growth, and malformed ear heads. Infection occurs at early seedling stage (10–30 days after sowing). Seed treatment with Metalaxyl provides early protection for about 25–30 days. Remove infected plants immediately to prevent spread. Continuous monitoring during first 40 days is essential.

Major pests include Stem Borer, Aphids, and Cutworms. Stem Borer damages stems during 20–40 days after sowing. Carbofuran application at early infestation reduces damage. Aphids suck sap and can be controlled using neem oil spray or recommended systemic insecticides. Cutworms damage seedlings at early stage and require soil treatment before sowing.

Nutrient deficiencies appear as yellowing leaves (nitrogen), stunted growth (phosphorus), leaf margin burn (potassium), and white streaks (zinc). Soil testing and micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done 90–120 days after sowing when ear heads turn brown and grains are hard and dry. Average grain yield ranges from 2–3 tons per hectare, while hybrids can produce 3–4 tons per hectare. Fodder yield is about 6–8 tons per hectare. After harvest, grains should be sun-dried to moisture below 12% and stored in airtight containers to prevent storage pests.

Organic ragi cultivation includes seed treatment with *Trichoderma*, farmyard manure application, neem cake incorporation, biofertilizers such as Azospirillum, and Panchagavya spray. Integrated disease and pest management combining resistant varieties, seed treatment, crop rotation, balanced fertilization, and need-based spraying reduces yield losses significantly.

Yield losses due to blast (30–50%), brown spot (15–25%), and stem borer (around 20%) can be minimized through early detection and timely intervention. Preventive sprays, if required, are generally repeated at 10–14 day intervals under favorable weather conditions.

Nutritionally, per 100 g, ragi contains approximately 72 g carbohydrates, 7–8 g protein, 344 mg calcium, 3–4 mg iron, and 11 g fiber. It strengthens bones, controls diabetes, improves digestion, prevents anemia, and supports heart health.

Modern technologies in ragi farming include high-yielding hybrids, climate-resilient varieties, integrated nutrient management, integrated pest management, and mechanized harvesting. Regionally, Karnataka practices transplanting methods, Tamil Nadu grows irrigated ragi, Uttarakhand cultivates it in hill regions, and Andhra Pradesh follows rainfed dryland farming.

Effective disease prevention in ragi requires monitoring from 15–70 days after sowing. Immediate action within one week of symptom appearance, proper seed treatment, balanced fertilization, and removal of infected plants significantly reduce disease spread and ensure stable production.

16. Barley

Barley, scientifically known as *Hordeum vulgare*, belongs to the Poaceae (grass) family and is one of the oldest cultivated cereal crops in the world, with a history of more than 10,000 years. In India, it is mainly grown during the Rabi season from October to March. Barley is used for food, livestock feed, and especially for malt production in breweries. Compared to wheat, barley is more tolerant to drought and salinity, making it suitable for marginal lands and dry regions. It is a short-duration crop, maturing in about 120–140 days, and plays an important role in food security and the malting industry.

Barley originated in the Fertile Crescent of the Middle East. It is an annual grass plant growing 60–120 cm tall, with long spikes bearing rows of grains. There are two main types: two-row barley, mainly used for malting, and six-row barley, primarily used for feed. It is a self-pollinated crop.

The crop prefers cool climatic conditions. Germination occurs best at 12–20°C, and growth is optimal between 15–25°C. It tolerates frost better than wheat but is sensitive to high temperatures during grain filling, which can reduce yield and grain quality. Barley requires 30–50 cm annual rainfall and performs well under dryland conditions. It is sensitive to waterlogging, which can promote root and leaf diseases.

Barley grows well in well-drained loam to sandy loam soils with a pH range of 6.0–8.5. It tolerates saline and alkaline soils better than wheat but should not be grown in heavy clay soils with poor drainage. Rajasthan is the largest barley-producing state in India, followed by Uttar Pradesh, Haryana, Madhya Pradesh, Punjab, and Bihar.

Popular varieties include RD 2035, RD 2552, BH 902, DWRUB 52, and RD 2660. Two-row types are preferred for malting quality.

Land preparation involves 2–3 ploughings to create a fine seedbed, followed by leveling and removal of weeds and residues. Sowing is done in October–November using a seed drill, which ensures uniform spacing of 22–25 cm between rows. The recommended seed rate is 75–100 kg per hectare. Seed treatment with Carbendazim or Thiram before sowing prevents seed-borne fungal diseases and protects seedlings for about 20–30 days after emergence.

Barley requires 2–3 irrigations depending on rainfall. Critical stages include tillering (25–30 days after sowing), booting (50–60 days), and grain filling (75–90 days). Over-irrigation should be avoided as it increases lodging and disease risk. Fertilizer recommendations include 40–60 kg nitrogen, 20–30 kg phosphorus, and 20 kg potassium per hectare. Nitrogen should be applied in two splits—basal and at first irrigation. Organic cultivation may include 5 tons of farmyard manure per hectare.

Weed competition is critical between 30–45 days after sowing. Hand weeding or herbicide such as Isoproturon helps control weeds effectively.

Among diseases, Stripe Rust caused by *Puccinia striiformis* is a major fungal disease that can cause 20–40% yield loss. It produces yellow stripes on leaves and reduces grain filling. It commonly appears from 40–70 days after sowing under cool, humid conditions. Prevention includes growing resistant varieties and balanced fertilization. At the first sign of symptoms, spray Propiconazole and repeat after 10–14 days if favorable weather persists. Monitoring should start from 35 days after sowing and continue until grain filling.

Leaf Blight caused by *Helminthosporium* species produces brown leaf spots and leaf drying, usually during 45–75 days after sowing. It can cause 15–30% yield loss. Spray Mancozeb at early symptom appearance and repeat after 10–12 days if needed. Proper crop rotation and field sanitation reduce infection.

Powdery Mildew caused by *Blumeria graminis* appears as white powdery growth on leaves under cool and humid conditions, generally during tillering to booting stages. Spray sulfur fungicide at first detection and repeat after 10 days if necessary. Balanced nitrogen use reduces severity.

Loose Smut caused by *Ustilago nuda* replaces grains with black powdery masses and is seed-borne. It can be completely prevented by seed treatment with Carboxin before sowing. Proper seed treatment protects the crop throughout the season.

Major pests include Aphids, Armyworms, and Termites. Aphids suck sap and may cause 10–20% yield loss if uncontrolled. Monitoring should begin from 40 days after sowing. Spray Imidacloprid at early infestation and repeat after 10 days if necessary. Neem oil can be used in organic systems. Armyworms feed on leaves and are controlled using Chlorantraniliprole. Termites damage roots, especially in dry soils, and can be managed through soil treatment before sowing or neem cake application.

Nutrient deficiencies appear as yellowing of older leaves (nitrogen), poor root growth (phosphorus), leaf margin burn (potassium), and white streaks (zinc). Balanced fertilization and zinc sulfate sprays correct deficiencies within 10–15 days.

Harvesting is done in March–April when spikes turn golden and grains are hard and dry. Harvesting can be manual or by combine harvester. Average yield ranges from 2–3 tons per hectare, while improved varieties can produce 4–5 tons per hectare under good management.

After harvest, grains should be dried to about 12% moisture and stored in a dry place to prevent storage pests. Barley is widely used in malt production for the beer industry, animal feed, flour, soups, breakfast cereals, and barley water.

Nutritionally, barley contains about 73 g carbohydrates, 10–12 g protein, 17 g fiber, and is rich in beta-glucan, which helps reduce cholesterol and control blood sugar. It supports heart health, improves digestion, and aids weight management.

Organic barley cultivation includes seed treatment with *Trichoderma*, application of farmyard manure, neem cake, biofertilizers, and organic pest control methods. Integrated disease and

pest management combining resistant varieties, seed treatment, crop rotation, balanced nutrition, and need-based spraying significantly reduces losses.

Yield losses from rust (20–40%), leaf blight (15–30%), and aphids (10–20%) can be minimized through early monitoring. Disease monitoring should be conducted every 7–10 days from 30–90 days after sowing. Fungicide sprays, if required, are generally repeated at 10–14 day intervals during favorable weather conditions.

Modern technologies in barley farming include high-yielding and climate-resilient varieties, precision nutrient management, mechanized harvesting, and integrated pest management (IPM). Regionally, Rajasthan practices rainfed barley cultivation, Haryana and Punjab focus on malting barley production, and Uttar Pradesh practices irrigated barley farming.

Effective disease prevention in barley depends on resistant varieties, proper seed treatment, balanced fertilization, timely irrigation, and close monitoring during the vegetative and reproductive stages. Immediate action within one week of symptom appearance greatly reduces disease spread and yield loss.

17. Groundnut

Groundnut, scientifically known as *Arachis hypogaea*, belongs to the Fabaceae (legume) family and is one of the most important oilseed crops in India. It is popularly called the “King of Oilseeds” because of its high oil content (45–50%). Groundnut is mainly cultivated during the Kharif season (June–September), and in some regions it is also grown during Rabi and Summer under irrigation. The crop is used for edible oil extraction, roasted nuts, peanut butter, sweets, snacks, and animal feed. The oil cake is a valuable cattle feed, and as a legume crop, groundnut improves soil fertility through nitrogen fixation. It is a short-duration crop maturing in 90–120 days and contributes significantly to export earnings.

Groundnut originated in South America (Brazil region) and is an annual herbaceous plant growing 30–50 cm tall. It produces yellow flowers and has a unique “pegging” process in which fertilized flowers produce pegs that penetrate the soil and develop pods underground. It has a deep taproot system and is self-pollinated.

The crop requires warm climatic conditions. Germination occurs best at 25–30°C, and optimal growth occurs at 20–35°C. It is sensitive to frost and extremely high temperatures during flowering, which can reduce pod formation. Groundnut requires 50–100 cm annual rainfall with good distribution. Waterlogging must be avoided because it promotes root and stem diseases. Bright sunshine is essential for good yield.

Groundnut grows best in sandy loam to loamy soils that are well-drained and loose, allowing proper peg penetration and pod development. The ideal pH range is 6.0–7.5. Heavy clay soils are unsuitable. Adequate calcium availability in soil is essential for proper pod formation. Gujarat is the largest producer, followed by Rajasthan, Tamil Nadu, Andhra Pradesh, Karnataka, Maharashtra, and Telangana.

There are two main types: bunch type and spreading type. Popular varieties include JL 24, GG 20, TMV 2, Kadiri 6, ICGV 91114, and TAG 24.

Land preparation requires 2–3 ploughings to create a fine, loose seedbed with proper leveling. Sowing is done in June–July (Kharif), October–November (Rabi), and January–February (Summer). The recommended seed rate is 100–125 kg per hectare with spacing of 30–45 cm between rows and 10–15 cm between plants. Seeds should be sown at 5–6 cm depth. Seed treatment with Thiram or Carbendazim prevents seed-borne diseases and protects seedlings for about 20–25 days. Rhizobium inoculation improves nitrogen fixation.

Irrigation is critical during flowering (30–40 days), pegging (40–50 days), and pod formation (50–80 days). Avoid water stagnation to prevent collar rot and stem rot. Fertilizer recommendations include 20–30 kg nitrogen, 40–60 kg phosphorus, and 40 kg potassium per hectare. Gypsum (calcium source) should be applied at flowering stage to improve pod filling. Zinc sulfate should be applied if deficiency appears. Organic farmers may apply 5 tons of farmyard manure per hectare.

Weed competition is severe during 15–45 days after sowing. Hand weeding or pre-emergence herbicide such as Pendimethalin helps control weeds effectively.

Among diseases, Tikka Disease (Leaf Spot) caused by *Cercospora arachidicola* produces brown circular spots and premature leaf drop, leading to 20–40% yield loss. It usually appears from 30–50 days after sowing, especially under humid conditions. Preventive measures include resistant varieties, balanced nutrition, and proper spacing. Spray Mancozeb at first symptom detection and repeat at 10–14 day intervals during favorable weather. Early spraying within 5–7 days of symptom appearance significantly reduces spread.

Rust caused by *Puccinia arachidis* produces brown pustules on leaves and may cause 15–30% yield loss. It appears during 45–75 days after sowing in humid weather. Spray Chlorothalonil or Tebuconazole at first detection and repeat after 10–12 days if required.

Collar Rot caused by *Aspergillus niger* affects seedlings, causing rotting near collar region and seedling death within 10–20 days after sowing. Seed treatment with fungicide and proper drainage prevent infection. Avoid deep sowing and waterlogging.

Stem Rot caused by *Sclerotium rolfsii* produces wilting and white fungal growth near soil surface, usually appearing after 40–60 days. Prevention includes crop rotation for 2–3 years, field sanitation, and proper drainage. Carbendazim spray or soil drenching at early detection helps control spread.

Bud Necrosis is a viral disease transmitted by thrips and can cause 10–50% yield loss. Symptoms include yellowing and stunted growth. Infection is severe if it occurs within 30–40 days after sowing. Control thrips population using Imidacloprid or neem oil sprays. Monitoring should begin from 15 days after emergence and sprays repeated every 10 days during heavy thrips activity.

Major pests include Aphids, Leaf Miner, White Grub, and Thrips. Aphids suck sap and can be controlled using Imidacloprid or neem oil. Leaf Miner damages leaves; Chlorantraniliprole spray at early infestation reduces damage. White Grubs damage roots and require soil treatment before sowing. Thrips transmit viral diseases and require regular monitoring and neem-based sprays in organic systems.

Nutrient deficiencies appear as pale leaves (nitrogen), stunted growth (phosphorus), leaf edge burn (potassium), poor pod formation (calcium), and small leaves (zinc). Soil testing and micronutrient sprays correct symptoms within 10–15 days.

Harvesting is done 90–120 days after sowing when leaves turn yellow and pods are mature and hard. Plants are uprooted manually or mechanically. Average yield ranges from 1.5–2.5 tons per hectare, while improved varieties can produce 3–4 tons per hectare.

After harvest, pods should be dried properly to reduce moisture to 8–10% before storage. Store in dry, ventilated conditions to prevent fungal growth and aflatoxin contamination. Oil extraction is done through cold pressing or solvent extraction methods.

Nutritionally, groundnut contains 25–30 g protein, 45–50 g fat, 16 g carbohydrates, and 8 g fiber per 100 g. It is rich in vitamin E and B-complex vitamins and provides healthy fats that support heart health and muscle strength.

Organic groundnut cultivation includes *Rhizobium* inoculation, neem cake application, *Trichoderma* seed treatment, Panchagavya sprays, and biofertilizers. Integrated Pest Management combining resistant varieties, crop rotation, seed treatment, balanced fertilization, and timely sprays significantly reduces yield losses.

Yield losses from leaf spot (20–40%), rust (15–30%), bud necrosis (10–50%), and white grub (around 20%) can be minimized through early monitoring. Field inspection should be done every 7–10 days from 20–80 days after sowing. Fungicide or insecticide sprays, if required, are generally repeated at 10–14 day intervals under favorable conditions.

Modern technologies in groundnut farming include high-yielding hybrids, drip irrigation, integrated nutrient management, mechanized harvesting, and climate-resilient varieties. Regionally, Gujarat practices large-scale commercial cultivation, Tamil Nadu grows irrigated groundnut, Andhra Pradesh follows rainfed farming, and Rajasthan promotes drought-tolerant varieties.

Effective disease prevention requires resistant varieties, proper seed treatment, balanced nutrition, gypsum application, good drainage, and continuous monitoring. Immediate action within one week of symptom detection significantly reduces disease spread and protects yield.

18. Mustard

Mustard, scientifically known as *Brassica juncea*, belongs to the Brassicaceae (crucifer) family and is one of the most important Rabi oilseed crops grown in India. It is cultivated mainly from October to March and plays a major role in edible oil production. Mustard seeds contain about 35–40% oil and 20–25% protein. The oil is widely used for cooking, pickles, traditional medicines, massage oils, and industrial products. Mustard oil cake serves as valuable cattle feed and organic manure. The tender leaves are consumed as a vegetable (sarson saag). India is among the leading producers globally, with Rajasthan contributing more than 40% of national production, followed by Uttar Pradesh, Haryana, Madhya Pradesh, West Bengal, and Punjab.

Mustard originated in Central Asia and is an annual herbaceous plant growing 1–2 meters tall. It bears bright yellow flowers and produces small round seeds that may be yellow, brown, or black. The crop is both self- and cross-pollinated. Major types include Indian mustard (*Brassica juncea*), Yellow mustard (*Sinapis alba*), Black mustard (*Brassica nigra*), Toria (early type), and Raya.

Mustard requires cool climatic conditions. Germination is best at 20–25°C, while vegetative growth occurs optimally at 10–25°C. Frost during flowering can damage flowers and reduce seed setting. High temperatures during maturity reduce oil content. The crop requires 25–40 cm of rainfall and performs well under dryland farming, but excess rain promotes fungal diseases. It grows best in well-drained loamy soils with pH 6.0–8.5 and tolerates slight alkalinity. Waterlogging must be strictly avoided.

Land preparation includes 2–3 ploughings to create a fine seedbed. Sowing is done in October–November using 4–6 kg seed per hectare with 30–45 cm row spacing and 10–15 cm plant spacing. Line sowing using a seed drill ensures uniform growth. Seed treatment with Thiram or Carbendazim protects seedlings for about 20–25 days against seed-borne pathogens. Azotobacter inoculation improves nitrogen availability.

Mustard requires 2–3 irrigations, mainly at flowering and pod formation stages. Avoid over-irrigation because excessive moisture increases disease incidence. Recommended fertilizer dose is 80–100 kg nitrogen, 40–60 kg phosphorus, and 40 kg potassium per hectare. Nitrogen should be applied in two splits—half at sowing and half at first irrigation. Sulfur (20–30 kg/ha) is essential for higher oil content and should be applied before sowing. Zinc sulfate is added if deficiency symptoms appear. Organic farmers apply 5–10 tons of farmyard manure.

Weed competition is critical during 20–40 days after sowing. Hand weeding or Pendimethalin herbicide application effectively controls weeds.

Among diseases, Alternaria blight caused by *Alternaria brassicae* is a major fungal disease causing dark circular spots on leaves and pods, leading to 20–40% yield loss. It usually appears 40–60 days after sowing under humid conditions. Prevention includes resistant varieties, balanced fertilization, and proper spacing to reduce humidity. At first symptom detection, spray

Mancozeb and repeat every 10–14 days if weather remains humid. Early spraying within 5–7 days of spotting symptoms prevents spread.

White rust caused by *Albugo candida* produces white pustules on leaves and swollen stems. It develops during cool, moist weather around 35–55 days after sowing. Seed treatment and spraying Metalaxyl at early symptom appearance control the disease. Repeat spray after 12–15 days if infection persists.

Downy mildew caused by *Peronospora parasitica* leads to yellow patches and fungal growth on the underside of leaves. It appears in cool and humid weather, usually 30–50 days after sowing. Maintaining proper plant spacing and good field drainage reduces risk. Fungicide spray at early detection, followed by another spray after 10–12 days, effectively controls it.

Powdery mildew caused by *Erysiphe cruciferarum* produces white powdery coating on leaves, commonly appearing late in the season (60–80 days after sowing) under dry and cool conditions. Sulfur spray controls it effectively. One to two sprays at 10-day intervals are usually sufficient.

Major pests include aphids, sawfly, and painted bug. Aphids are the most destructive and can cause 10–50% yield loss by sucking sap during flowering and pod stages (45–75 days after sowing). Early detection is essential. If aphid colonies appear, spray Imidacloprid immediately and repeat after 10 days if needed. Neem oil sprays at 7-day intervals help in organic systems. Sawfly larvae feed on leaves in early crop stages (20–35 days). Malathion spray controls heavy infestation. Painted bugs suck sap from pods and can be controlled by Chlorpyrifos spray when infestation crosses threshold.

Nutrient deficiencies appear as yellowing leaves (nitrogen), poor flowering (phosphorus), leaf margin burn (potassium), pale young leaves (sulfur), and stunted growth (zinc). Balanced fertilization and micronutrient sprays correct deficiencies within 10–15 days.

Harvesting is done 110–140 days after sowing when pods turn yellow-brown and seeds become hard. Delayed harvesting causes shattering losses. Average yield ranges from 1.5–2 tons per hectare, while hybrids yield 2.5–3 tons per hectare.

After harvest, seeds must be dried to 8–10% moisture before storage to prevent fungal growth. Oil extraction is done through cold pressing or expeller methods. Mustard oil cake is used as fertilizer and feed, and mustard oil is also used in biodiesel production.

Nutritionally, mustard seeds contain 20–25 g protein, 35–40 g fat, and are rich in omega-3 fatty acids. They support heart health, digestion, and possess anti-inflammatory and antioxidant properties.

Organic mustard cultivation includes Trichoderma seed treatment, neem cake application, Panchagavya sprays, biofertilizers, and neem-based pest control. Integrated Pest Management combining resistant varieties, timely sowing, balanced nutrition, and need-based spraying significantly reduces yield loss.

Yield losses may reach 20–40% from *Alternaria* blight, 15–30% from white rust, and up to 50% from aphids if unmanaged. Regular field monitoring every 7–10 days from 25 to 80 days after sowing is critical. Immediate action within one week of symptom appearance, along with timely repeat sprays at 10–14 day intervals, effectively suppresses disease spread.

Modern technologies include hybrid varieties, precision nutrient management, integrated pest management, mechanized harvesting, and climate-resilient cultivars. Regionally, Rajasthan practices large-scale rainfed cultivation, Haryana and Punjab focus on irrigated mustard, Uttar Pradesh uses mixed cropping systems, and West Bengal grows early toria varieties.

Effective disease prevention depends on resistant varieties, proper seed treatment, balanced fertilization (especially sulfur), good drainage, timely irrigation, and regular monitoring. Early intervention within 5–7 days of detection and maintaining 10–14 day protective intervals during favorable weather ensures minimal crop loss and maximum yield stability.

19. Soybean

Soybean (*Glycine max*) is an important oilseed and pulse crop belonging to the Fabaceae family and is popularly called the “Golden Bean” due to its high protein (38–42%) and oil (18–22%) content. In India, it is mainly cultivated during the Kharif season (June–September) under rainfed conditions. Soybean plays a crucial role in edible oil production, livestock and poultry feed industries, and export markets. It is also used in soy milk, tofu, soy flour, textured vegetable protein, and biodiesel production. As a leguminous crop, it fixes atmospheric nitrogen through Rhizobium nodules, thereby improving soil fertility and benefiting subsequent crops.

Soybean originated in China and is an annual herbaceous plant growing 30–100 cm tall with hairy stems and trifoliate leaves. The pods contain 2–4 seeds and develop above ground. It is a self-pollinated, short-day plant, sensitive to photoperiod. The crop requires warm climatic conditions, with optimal germination at 20–30°C and growth at 25–30°C. Frost damages the crop severely, and high temperatures during flowering reduce pod setting. It requires 60–100 cm of well-distributed rainfall and is highly sensitive to waterlogging, especially during flowering and pod formation.

Soybean grows best in well-drained loamy soils with pH 6.0–7.5. Heavy clay or acidic soils and stagnant water conditions should be avoided. In India, Madhya Pradesh is the largest producer and is known as the “Soybean State of India,” followed by Maharashtra, Rajasthan, Karnataka, Telangana, and Uttar Pradesh.

Land preparation includes 2–3 ploughings to obtain fine tilth and ensure proper drainage. Sowing is done at the onset of monsoon (June–July) using 60–80 kg seed per hectare, spaced 30–45 cm between rows and 5–7 cm between plants. Seed treatment with Thiram or Carbendazim protects seedlings from seed-borne diseases for about 20–25 days after sowing. Rhizobium inoculation enhances nitrogen fixation and early vigor.

Soybean is mainly rainfed but requires moisture during flowering, pod formation, and seed filling stages. Excess water should be avoided because prolonged waterlogging for more than 3–4 days can lead to root diseases.

The recommended fertilizer dose is 20–30 kg nitrogen, 60–80 kg phosphorus, and 40 kg potassium per hectare. Although soybean fixes nitrogen, a small starter dose improves early growth. Sulfur enhances oil content, and zinc sulfate is applied if deficiency appears. Farmyard manure (5 tons/ha) improves soil health and microbial activity.

Weed competition during 15–45 days after sowing can reduce yield drastically. Hand weeding or pre-emergence application of Pendimethalin is effective. Two weeding at 20 and 35 days are generally sufficient.

Among major diseases, Soybean Rust caused by *Phakopsora pachyrhizi* is highly destructive and can cause 30–70% yield loss. It usually appears 35–60 days after sowing under warm and

humid conditions. Symptoms include brown pustules on leaves and premature leaf drop. Preventive measures include resistant varieties, early sowing, and balanced fertilization. At the first appearance of symptoms, spray Propiconazole or Tebuconazole. Repeat spraying after 10–14 days if humidity persists. Early intervention within 5–7 days of symptom detection prevents severe spread.

Yellow Mosaic Virus (YMV), transmitted by whiteflies, causes yellow patches and stunted growth, leading to 20–60% yield loss. It typically appears 25–45 days after sowing. Prevention involves controlling whitefly populations through Imidacloprid spray or neem-based insecticides. Seed treatment with systemic insecticides provides protection for the first 20–25 days. Monitoring fields every 7 days during vegetative stage helps in early detection. Once infected, plants cannot be cured, so removing infected plants immediately helps limit spread.

Rhizoctonia Root Rot caused by *Rhizoctonia solani* leads to root decay and wilting, particularly under poorly drained conditions. It often occurs within 20–40 days after sowing. Preventive measures include proper drainage, crop rotation for 2–3 years, and seed treatment with fungicide or Trichoderma. Avoid irrigation for at least 7–10 days after heavy rainfall if soil remains wet.

Anthrachnose caused by *Colletotrichum truncatum* affects pods and seeds, producing dark lesions and reducing seed quality. It usually appears during pod development (50–75 days after sowing). Mancozeb spray at early symptom detection, repeated after 10–12 days, effectively controls the disease.

Major insect pests include stem fly, whitefly, leaf-eating caterpillar, and pod borer. Stem fly attacks early seedlings (10–25 days), causing stem tunneling; Imidacloprid seed treatment protects seedlings for about 20–25 days. Whiteflies transmit viral diseases and require regular monitoring. Leaf-eating caterpillars cause defoliation during vegetative stage; spraying Chlorantraniliprole at threshold levels controls them. Pod borers damage pods during reproductive stage (45–80 days); pheromone traps and need-based spraying prevent heavy losses.

Nutrient deficiencies appear as yellow leaves (nitrogen), stunted growth (phosphorus), leaf edge burn (potassium), pale young leaves (sulfur), and small leaves (zinc). Correction through balanced fertilization and foliar micronutrient spray shows improvement within 10–15 days.

Soybean is ready for harvest 90–110 days after sowing when leaves turn yellow and pods become brown and dry. Mechanical harvesting reduces losses. Average yield ranges from 1.5–2.5 tons per hectare, while improved varieties under good management yield 3–4 tons per hectare.

After harvest, seeds must be dried to 10–12% moisture to prevent fungal growth. Oil extraction is done through solvent extraction methods, and oil cake serves as protein-rich livestock feed.

Nutritionally, soybean contains about 40 g protein, 20 g fat, and 9 g fiber per 100 g, and is rich in isoflavones. It supports heart health, reduces cholesterol, aids muscle building, and is highly beneficial for vegetarians.

Organic soybean cultivation includes *Rhizobium* inoculation, neem cake application, *Trichoderma* seed treatment, biofertilizers, and neem-based pest management. Integrated Pest Management combining resistant varieties, timely sowing, crop rotation, balanced nutrition, and regular field inspection every 7–10 days significantly reduces disease incidence.

Yield losses can reach 30–70% from rust, 20–60% from viral diseases, and 15–25% from pod borers if unmanaged. Preventive measures such as seed treatment before sowing, early monitoring starting from 15 days after sowing, and timely spraying at 10–14 day intervals during favorable weather are essential to avoid severe outbreaks. Immediate control within one week of symptom detection and maintaining a disease-free field environment for 2–3 weeks after treatment ensures effective management and stable yield.

Modern soybean farming uses high-yielding varieties, precision nutrient management, mechanized harvesting, integrated pest management, and climate-resilient cultivars. Regionally, Madhya Pradesh practices large-scale rainfed soybean cultivation, Maharashtra adopts intercropping with cotton, Rajasthan follows mixed cropping systems, and Karnataka grows irrigated soybean.

Effective disease prevention depends on early detection, seed treatment protection for the first 20–25 days, maintaining proper drainage throughout the crop period, and repeating protective sprays every 10–14 days during high-risk periods to prevent disease recurrence and ensure maximum productivity.

20. Sunflower

Sunflower (*Helianthus annuus*) is an important oilseed crop belonging to the Asteraceae family and is cultivated widely in India and across the world. It is valued for its high-quality edible oil containing 38–45% oil with a high proportion of unsaturated fatty acids and rich Vitamin E content. Because of its short duration of 85–110 days and adaptability to Kharif, Rabi, and Summer seasons, sunflower is considered a versatile crop suitable for different agro-climatic conditions. The oil is widely used for cooking and processed foods, while oil cake serves as nutritious cattle feed. It is also an excellent crop for rotation because of its deep root system and moderate nutrient requirement.

Sunflower originated in North America and is an annual herbaceous plant growing 1–3 meters tall. It produces a large yellow flower head called a capitulum, and seeds develop in the center of this head. It is a cross-pollinated crop mainly dependent on insects like bees for pollination. The plant has a deep taproot system that helps it tolerate moderate drought conditions.

The crop grows best at 20–25°C for germination and 25–30°C during vegetative growth. It tolerates moderate heat but is sensitive to frost. It requires 50–75 cm rainfall annually and performs well in both irrigated and rainfed conditions. However, waterlogging for more than 2–3 days can severely affect root health and increase fungal disease incidence. Sunflower requires full sunlight and is a day-neutral plant.

It thrives in well-drained loamy soils with pH between 6.5 and 8.0. Heavy clay soils with poor drainage should be avoided. In India, Karnataka is the largest producer, followed by Andhra Pradesh, Telangana, Maharashtra, Tamil Nadu, and Haryana.

Land preparation involves 2–3 ploughings to create a fine seedbed and ensure good drainage. Sowing is done in June–July (Kharif), September–October (Rabi), or January–February (Summer). Seed rate is 5–7 kg per hectare with spacing of 45–60 cm between rows and 20–30 cm between plants. Seeds should be treated with Thiram or Carbendazim to protect against early seedling diseases for about 20–25 days after sowing.

Critical irrigation stages include flower initiation (35–40 days after sowing), flowering (45–55 days), and seed filling (60–75 days). Excess irrigation must be avoided to prevent root rot and downy mildew. Recommended fertilizer dose includes 60–80 kg nitrogen, 60 kg phosphorus, and 40 kg potassium per hectare. Nitrogen should be applied in two splits—half at sowing and half at 30 days after sowing. Boron is essential for proper seed set, and zinc sulfate should be applied if deficiency symptoms appear. Application of 5 tons farmyard manure improves soil fertility.

Weed competition during 20–40 days after sowing is critical. Hand weeding or Pendimethalin application effectively controls weeds and protects yield.

Among diseases, Alternaria leaf spot caused by *Alternaria helianthi* is common, appearing 30–50 days after sowing in humid weather. It produces brown circular spots leading to leaf drying and 20–40% yield loss. Preventive measures include resistant varieties, proper spacing for air circulation, and balanced fertilization. At first symptom detection, spray Mancozeb and repeat after 10–14 days if humidity persists. Early spraying within 5–7 days prevents severe spread.

Downy mildew caused by *Plasmopara halstedii* affects young plants, usually 15–30 days after sowing, especially under cool and wet conditions. Symptoms include yellowing and stunted growth. Seed treatment with Metalaxyl protects seedlings for about 20–25 days. Infected plants should be removed immediately to prevent spread. Avoid irrigation for 7–10 days if soil remains excessively wet.

Rust caused by *Puccinia helianthi* appears 45–70 days after sowing as brown pustules on leaves. It spreads rapidly under warm and humid conditions. Propiconazole spray at early detection followed by another spray after 12–14 days effectively controls the disease.

Powdery mildew caused by *Erysiphe cichoracearum* appears late in the crop (60–80 days after sowing) as white powdery coating. It is favored by dry weather with cool nights. Sulfur spray once or twice at 10-day intervals controls it effectively.

Major insect pests include Helicoverpa (head borer), aphids, and whiteflies. Head borer attacks during flowering and seed formation (45–75 days after sowing), causing 20–50% yield loss. Pheromone traps should be installed early, and Chlorantraniliprole spray applied when larvae are small. Aphids suck sap during vegetative and flowering stages and can be controlled by Imidacloprid or neem oil sprays at 7–10 day intervals. Whiteflies also suck sap and transmit viral diseases; early monitoring and insecticide spray prevent spread.

Nutrient deficiencies include yellowing leaves (nitrogen), poor root growth (phosphorus), leaf margin burn (potassium), poor seed set (boron), and small leaves (zinc). Corrective foliar sprays show visible improvement within 10–15 days.

Sunflower is ready for harvest 85–110 days after sowing when the back of the flower head turns yellow and seeds become hard. Delayed harvest increases bird damage and shattering losses. Average yield ranges from 1.5–2 tons per hectare, while hybrids yield 2.5–3 tons per hectare.

After harvest, seeds should be dried to 9–10% moisture before storage to prevent fungal growth. Oil extraction is done through cold press or solvent methods. Sunflower oil cake is used as protein-rich cattle feed.

Nutritionally, sunflower seeds contain about 45 g fat, 20 g protein, and high Vitamin E per 100 g. They help reduce cholesterol, support heart health, improve skin condition, and boost immunity due to antioxidant properties.

Organic cultivation involves Trichoderma seed treatment, neem cake application, biofertilizers, and neem-based pest control. Integrated Pest Management combining resistant hybrids, crop

rotation, balanced fertilization, proper spacing, and weekly monitoring from 20 to 80 days after sowing significantly reduces disease outbreaks.

Yield losses may reach 20–40% from *Alternaria*, 30% from downy mildew, and up to 50% from head borer if unmanaged. Regular field inspection every 7–10 days, immediate fungicide or insecticide application within one week of symptom appearance, and maintaining protective spray intervals of 10–14 days during favorable weather effectively prevent major damage.

Modern sunflower farming includes hybrid seed technology, drip irrigation, precision nutrient management, mechanized harvesting, and climate-resilient cultivars. Regionally, Karnataka practices rainfed cultivation, Andhra Pradesh grows irrigated sunflower, Maharashtra adopts intercropping with pulses, and Tamil Nadu promotes summer sunflower farming.

Effective prevention depends on timely seed treatment (protection for first 20–25 days), good drainage to avoid prolonged wet soil conditions, early detection from 15 days after sowing, and repeat protective sprays every 10–14 days during high-risk periods to maintain disease-free growth and ensure maximum yield.

Sesame

Sesame (*Sesamum indicum*), belonging to the Pedaliaceae family, is one of the oldest oilseed crops cultivated in the world and holds significant importance in India. It is mainly grown during the Kharif season (June–September), and in some regions during Rabi and Summer under irrigation. Sesame seeds contain about 45–55% oil and 18–25% protein, making them highly valuable for edible oil production, sweets, bakery products, and traditional medicines. Sesame oil is widely used in cooking and Ayurvedic applications due to its stability and antioxidant content. The oil cake serves as livestock feed and organic fertilizer. The crop is drought-tolerant and suitable for dryland farming, making it important for semi-arid agriculture and export markets.

Sesame originated in Africa and is one of the earliest domesticated oilseed crops. It is an annual herbaceous plant growing 50–150 cm tall with bell-shaped white or pink flowers. The fruit is a capsule containing numerous small seeds. It has a deep taproot system that enables it to withstand drought conditions. Sesame is mainly self-pollinated.

There are three main types: white sesame (preferred for export), black sesame (used in medicines), and brown sesame. The crop requires warm climatic conditions, with optimal germination at 25–30°C and growth at 25–35°C. It is sensitive to frost and heavy rainfall. Excess moisture for more than 2–3 consecutive days may cause root and stem diseases. It requires 50–100 cm rainfall annually but performs better in dry conditions.

Sesame grows best in well-drained sandy loam soils with pH 5.5–8.0. It tolerates poor soils but cannot survive waterlogged conditions. In India, major sesame-growing states include Uttar Pradesh, Rajasthan, Gujarat, West Bengal, Madhya Pradesh, Andhra Pradesh, Telangana, and Tamil Nadu. India is one of the largest producers and exporters globally.

Land preparation requires 2–3 ploughings to obtain a fine seedbed with good drainage. Sowing is done in June–July (Kharif), October (Rabi), or February (Summer). The seed rate is 3–5 kg per hectare with spacing of 30–45 cm between rows and 10–15 cm between plants. Seeds should be treated with Thiram or Carbendazim before sowing to protect seedlings for 20–25 days against seed-borne fungi. Organic farmers use Trichoderma for protection.

Sesame is mainly rainfed but needs adequate moisture during flowering and capsule formation (35–60 days after sowing). Water stagnation must be avoided, especially during early growth and flowering stages.

The recommended fertilizer dose is 40–60 kg nitrogen, 20–30 kg phosphorus, and 20 kg potassium per hectare. Sulfur improves oil content, and zinc sulfate is applied if deficiency appears. Application of 5 tons farmyard manure improves soil fertility and moisture retention.

Weed competition is critical during the first 15–30 days after sowing. Two hand weedings at 15 and 30 days effectively control weeds. Pre-emergence herbicide application also reduces competition.

Among diseases, Phyllody is one of the most destructive, caused by a mycoplasma-like organism and transmitted by leafhoppers. It appears 30–50 days after sowing. Symptoms include conversion of floral parts into leafy structures, leading to sterile plants and 30–70% yield loss. There is no direct cure once infected. Prevention includes controlling leafhopper vectors early. Spray Imidacloprid when leafhoppers are first observed and repeat after 10–12 days if infestation continues. Regular monitoring from 20 days after sowing at 7-day intervals is essential. Remove infected plants immediately to prevent spread.

Alternaria leaf spot caused by *Alternaria sesami* produces brown spots on leaves and leads to premature leaf fall, usually appearing 35–60 days after sowing under humid conditions. Preventive measures include resistant varieties, proper spacing, and balanced fertilization. Spray Mancozeb at the first appearance of spots and repeat after 10–14 days if weather remains humid. Early spraying within 5–7 days of symptom detection prevents severe yield loss.

Stem rot caused by *Macrophomina phaseolina* appears during hot and dry conditions, usually 40–70 days after sowing. Symptoms include wilting and black lesions on the stem base. Prevention includes crop rotation for 2–3 years, proper drainage, and seed treatment before sowing. Avoid irrigation for at least 7–10 days if soil is excessively wet.

Powdery mildew caused by *Oidium sesami* appears late in the crop (60–80 days after sowing) as white powdery growth on leaves. Sulfur spray controls it effectively. One or two sprays at 10-day intervals are usually sufficient.

Major insect pests include leaf roller, capsule borer, and aphids. Leaf roller damages leaves during vegetative stage (20–40 days). Neem oil spray at 7–10 day intervals controls infestation. Capsule borer attacks developing capsules (45–75 days), reducing seed yield by 15–25%. Pheromone traps and Chlorantraniliprole spray when larvae are small are effective. Aphids suck sap and weaken plants; Imidacloprid spray controls severe infestation.

Nutrient deficiencies include yellowing leaves (nitrogen), poor growth (phosphorus), leaf margin burn (potassium), stunted growth (zinc), and pale young leaves (sulfur). Balanced fertilization and micronutrient sprays correct symptoms within 10–15 days.

Sesame is ready for harvest 80–100 days after sowing when capsules turn yellow. Harvesting must be done before capsule shattering. Plants are cut and dried before threshing. Average yield ranges from 0.6–1.2 tons per hectare, while improved varieties can yield up to 1.5 tons per hectare.

After harvest, capsules should be dried properly and moisture reduced to 6–8% to prevent fungal contamination. Sesame oil is extracted through cold press methods, and oil cake is used as fertilizer and feed.

Nutritionally, sesame seeds contain about 20 g protein, 50 g fat, 975 mg calcium, and 14 mg iron per 100 g. They are rich in antioxidants such as sesamin and sesamol. Health benefits include improved heart health, stronger bones, better digestion, blood pressure control, and enhanced immunity.

Organic sesame cultivation involves Trichoderma seed treatment, neem cake application, biofertilizers, and neem-based pest management. Integrated Pest Management combining resistant varieties, early sowing, balanced nutrients, proper drainage, and weekly field monitoring significantly reduces yield loss.

Yield losses may reach 30–70% from phyllody, 20–40% from *Alternaria*, and 15–25% from capsule borer if unmanaged. Early seed treatment protects seedlings for 20–25 days. Regular inspection starting 15–20 days after sowing and immediate control within one week of symptom appearance are essential. Repeating protective sprays every 10–14 days during favorable weather conditions ensures effective disease suppression and stable yield.

Modern sesame farming includes high-yielding varieties, Integrated Nutrient Management, Integrated Pest Management, mechanized harvesting, and climate-resilient cultivars. Regionally, Uttar Pradesh grows rainfed sesame, Gujarat produces export-quality white sesame, Tamil Nadu promotes summer sesame, and West Bengal practices mixed cropping systems.

Effective prevention depends on proper seed treatment before sowing, maintaining well-drained soil, avoiding water stagnation for more than 2–3 days, and implementing timely pest and disease control measures within 5–7 days of detection to maintain a healthy crop throughout its 80–100 day growth period.